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# Reliable Causal Estimation under Latent Markov Confounding

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## Abstract

Causal inference from dependent observational time series is complicated when an unobserved, persistent state jointly affects treatment assignment and outcomes. Standard Double Machine Learning (DML) controls for observed covariates through orthogonal residualization, but does not by itself resolve confounding induced by an imperfect proxy for latent state dynamics. We study causal estimation under latent Markov confounding and characterize how latent-state uncertainty and downstream task conditioning propagate into causal error. We develop a regime-aware orthogonal estimator combining probabilistic state inference, temporally purged cross-fitting, and regularized inversion of the cross-regime score Jacobian. Our analysis separates nuisance-estimation error, latent-state recovery error, and variability induced by persistent regime occupation, and gives conditions under which the estimator approaches the corresponding oracle-state estimator as latent-state uncertainty decreases. We evaluate these properties using real-data-calibrated semi-synthetic time series constructed from high-frequency atmospheric observations across heterogeneous Indian cities, preserving persistence, seasonality, missingness, and regime overlap while keeping causal effects known. Across controlled variations in state persistence and observability, the experiments show how latent-state uncertainty predicts causal bias and when regime-aware estimation improves over standard and temporally blocked DML. Real sensor networks provide an observational stress test across four megacities spanning distinct

residual treatment variance scales and telemetry artifacts; falsification checks are reported explicitly where temporal causal interpretation is unsupported. Together, the results show that reliable causal estimation with latent sequential confounding requires evaluating representation uncertainty jointly with the information geometry of the downstream task.

## 1 INTRODUCTION

Many observational time series are driven by a persistent state that nobody measures: the boundary-layer regime of an airshed, the liquidity regime of a market, the operating mode of a plant. When that state moves both a treatment and an outcome, adjusting for observed covariates does not identify the effect. The standard remedy is a two-stage recipe: infer the state with a hidden Markov model (HMM) from auxiliary proxies, then estimate regime-aware effects with Double Machine Learning (DML; Chernozhukov et al., 2018). The recipe comes with an implicit reliability rule: *trust the estimate when the posterior is confident and the score Jacobian is well conditioned*. Both quantities are routinely reported, and both appear in error bounds of the form  $C\varepsilon_\gamma/\lambda_{\min}(\mathbf{J})$ , which we also prove (Theorem 3).

This paper shows that the rule fails in a specific and checkable way. A *generated* state does not only add noise to the downstream estimating equation; it changes that equation’s geometry. Mixing regimes leaks between-regime treatment contrasts into the residualized treatment, so the smallest eigenvalue of the generated-state Gram matrix can *grow* as the state estimate gets worse. In a controlled two-regime design it grows 70× while the causal error first rises and then falls (Figure 1a–b), and with posteriors learned by EM the generated-state matrix looks better conditioned than the oracle-state matrix in 82% of runs (Figure 1c). We call this *phantom causal resolution*: the estimate becomes less trustworthy while its standard conditioning diagnostic becomes more reassuring.

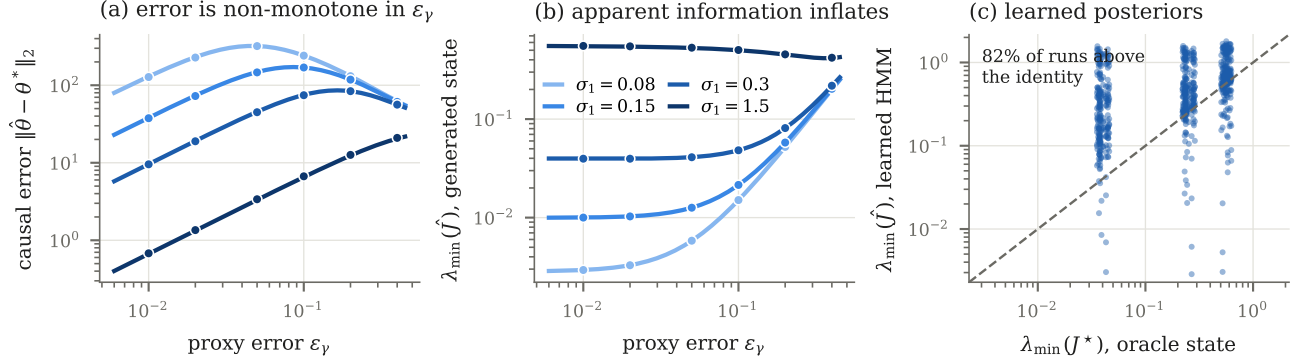


Figure 1: **Phantom causal resolution.** (a) Coupled regime-score error against proxy error  $\varepsilon_\gamma$  for four treatment-innovation scales  $\sigma_1$  (dots: median of 100 runs of the  $7 \times 7$  factorial design; lines: the exact population law of Proposition 1, with no fitted parameters). Error is non-monotone in  $\varepsilon_\gamma$ , peaking near  $\varepsilon^* \approx 2\sigma/|\Delta_m|$ . (b) The smallest eigenvalue of the generated-state Gram matrix in the same cells: unless  $\sigma_1$  is large, it grows with proxy error ( $70\times$  at  $\sigma_1 = 0.08$ ), so a worse state estimate looks *better* conditioned. (c) With posteriors learned by EM from noisy proxies (720 runs),  $\lambda_{\min}(\hat{J})$  exceeds the oracle-state value in 82% of runs (median ratio 1.9, 90th percentile 11).

The mechanism has a closed form, and the closed form says how to build the estimator. For the coupled regime score we derive the exact population displacement under state confusion (Proposition 1). It reproduces all 42 cells of a 4,900-run factorial experiment with observed/predicted ratio 0.999 (interquartile range 0.993–1.008), and it locates the damage in the nuisance step: *weighting* regime nuisances by the posterior leaks regime contrasts into the score. *Conditioning* the nuisances on the posterior instead makes the coupled score exactly Neyman-orthogonal for every posterior (Lemma 1). What remains is an exact omitted-variable identity in the *residual state variance*  $v$  that the posterior leaves behind (Theorem 4). For a pooled effect this is  $\Delta_T \Delta_g v / (\Delta_T^2 v + \sigma^2)$ ; regime effects obey the same law in matrix form. The conditioning diagnostic shares a factor with the bias numerator, so no function of  $(\varepsilon_\gamma, \lambda_{\min})$  alone can certify reliability. What is needed instead is the residual state information and the regime shifts it multiplies. Both are estimable up to one unidentified outcome shift, which supports a sensitivity analysis.

**Contributions.** (i) *Failure and limits:* an exact decomposition of the omitted-regime bias of standard DML, including bias amplification, and non-identification when proxies carry no state information (Theorems 1–2). (ii) *Mechanism:* exact laws for how state confusion enters coupled and posterior-conditioned estimators. They explain phantom resolution, the non-monotonicity of error, and why a posterior helps as an adjustment feature but hurts as a nuisance weight (Proposition 1, Theorem 4). (iii) *Estimator:* overlap-aware regime DML (OR-DML) with posterior-adjusted nuisances, purged block cross-fitting

and spectral regularization. It comes with exact orthogonality, a degradation bound separating nuisance from state-recovery error, and asymptotic normality including HMM-parameter and regime-occupation variance (Lemma 1, Theorems 3, 5). Across 900 synthetic datasets it is the most accurate regime-aware estimator, up to ties, at every proxy separation and persistence level we test. The posterior-weighted design of earlier drafts loses to hard regime fixed effects and is reported as an ablation. (iv) *Evidence:* a factorial design verified against the closed form; learned-posterior experiments that remove its circularity; a real-data-calibrated semi-synthetic benchmark on four Indian cities with real gaps and calibrated persistence and overlap; and an audited sensor stress test with aligned placebos and robustness values.

**Related work.** Orthogonal and debiased estimation (Chernozhukov et al., 2018, 2022b; Foster and Syrgkanis, 2023; Nie and Wager, 2021) and its time-series extensions (Lewis and Syrgkanis, 2020; Semenova et al., 2023; Ciganovic et al., 2026) assume the confounders are observed. Sequential deconfounders learn latent confounders with neural representations (Bica et al., 2020; Hatt and Feuerriegel, 2021); our results caution that representation quality and downstream reliability are different objects. Latent-state and proximal approaches target identification (Miao et al., 2018; Tchetgen Tchetgen et al., 2020; Mastouri et al., 2021; Cui et al., 2024; Clouth et al., 2025), and regime-switching models go back to Hamilton (1989); we instead quantify what a *plug-in* posterior leaves behind. Our bias law is an omitted-variable identity (Greenland, 1980; Cinelli and Hazlett, 2020; Chernozhukov et al., 2022), and its non-monotonicity is the algebra of bias amplification

(Pearl, 2010; Ding et al., 2017); we claim neither as new. What is new is its calibrated, dependent-data and regime-coupled form, the exact law for regime-weighted scores, the orthogonal design that follows from them, and the consequence that the Gram-matrix diagnostics used for weak overlap (Crump et al., 2009; D’Amour et al., 2021) can move in the wrong direction when the state is generated. Posterior-based adjustment is known to fail without identification (Wang and Blei, 2019; D’Amour, 2019); we characterize how it fails.

## 2 SETTING AND THE FAILURE OF STANDARD DML

We observe  $(Y_t, T_t, X_t, Z_t)_{t=1}^N$  generated by

$$S_t \sim \text{Markov}(\Pi, \pi_0), \quad Z_t = \mu_Z(S_t) + \xi_t, \quad (1)$$

$$T_t = m(X_t, S_t) + V_t, \quad Y_t = \theta(S_t)T_t + g(X_t, S_t) + U_t, \quad (2)$$

with  $S_t \in \{1, \dots, K\}$  latent,  $\mathbb{E}[V_t | X_t, S_t] = 0$ ,  $\mathbb{E}[U_t | X_t, S_t, T_t] = 0$  and  $\theta(S_t) = \sum_k \theta_k^* \mathbb{I}(S_t = k)$ . Targets are the regime effects  $\theta_k^*$  and the stationary average  $\theta_{\text{ATE}}^* = \sum_k \pi_k^* \theta_k^*$ . In our application  $T$  is  $\text{NO}_2$ ,  $Y$  is  $\text{PM}_{2.5}$ ,  $X$  collects lagged pollutants and weather, and  $Z$  is meteorology. Regime-conditional unconfoundedness, overlap and positivity, bounded variables, geometric  $\alpha$ -mixing and the usual  $\mathcal{O}_P(N^{-1/4})$  nuisance rates are stated as Assumptions 1–4 in Appendix A.

**Theorem 1** (Omitted regime bias). *Standard DML that residualizes on  $X_t$  alone satisfies  $\hat{\theta}_{\text{naive}} \rightarrow_p \sum_k \omega_k \theta_k^* + \mathcal{B}_{\text{het}} + \mathcal{B}_{\text{conf}}$ , with  $\omega_k = \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2] / \mathbb{E}[\tilde{T}_t^2]$  and  $\mathcal{B}_{\text{conf}} = \sum_k \mathbb{E}[\pi_k(X) \Delta m_k(X) \Delta g_k(X)] / \mathbb{E}[\tilde{T}_t^2]$ , where  $\Delta m_k, \Delta g_k$  are regime deviations of the treatment mean and outcome baseline (full statement and proof in Appendix B).*

Because  $\mathbb{E}[\tilde{T}_t^2]$  shrinks as  $X$  explains more of  $T$  while the latent covariance in the numerator does not, residualizing on more observed controls can *amplify* omitted-regime bias and flip its sign (Pearl, 2010). In our synthetic frontier, standard DML has bias 8.44 on a true ATE of 1.53 at every level of proxy quality.

**Theorem 2** (Non-identification without state information). *If  $P(Z_t | S_t = j) = P(Z_t | S_t = k)$  for all  $j, k$  and transitions are memoryless, there exist two models satisfying all assumptions, with identical laws of  $(Y, T, X, Z)_{1:N}$  for every  $N$  and distinct, non-permuted effect vectors (Appendix C).*

Between these extremes lies the regime that matters in practice: proxies that are informative but imperfect. The rest of the paper is about that regime.

## 3 OVERLAP-AWARE REGIME DML

Stepwise latent-model analysis is established, and classification error already motivates explicit corrections (Di Mari et al., 2016). We test the natural two-stage heuristic of posterior-weighted statewide nuisance regressions, rather than joint or bias-corrected latent-model likelihoods. Proposition 1 isolates soft target weighting with oracle regime nuisances; Lemma 1 addresses nuisance estimation; Theorem 4 characterizes the residual bias after posterior adjustment.

OR-DML infers posteriors  $\gamma_{tk} = P(S_t = k | Z_{1:N})$  (smoothing) or  $P(S_t = k | Z_{1:t})$  (causal filtering) with a Gaussian HMM. It cross-fits one pair of nuisances on the posterior-augmented features  $F_t = (X_t, \gamma_t, \gamma_t \otimes X_t)$ ,  $\hat{m}(F) \approx \mathbb{E}[T | F]$  and  $\hat{\mu}(F) \approx \mathbb{E}[Y | F]$ , on contiguous blocks separated by an embargo of  $\tau^*$  steps. With  $\tilde{T}_t = T_t - \hat{m}(F_t)$  and  $\tilde{Y}_t = Y_t - \hat{\mu}(F_t)$  it solves the coupled system

$$\hat{J}_{jk} = \frac{1}{N} \sum_t \gamma_{tj} \gamma_{tk} \tilde{T}_t^2, \quad \hat{S}_k = \frac{1}{N} \sum_t \gamma_{tk} \tilde{T}_t \tilde{Y}_t, \quad (3)$$

$\hat{\theta}_\lambda = (\hat{J} + \lambda I)^{-1} \hat{S}$ . The posterior enters twice: as an *adjustment feature* in the nuisances (fine resolution) and as a *weight* that defines the regime effects (coarse target). Bartlett bands guide the finite-lag embargo search; they do not estimate  $c_\beta$  or certify independence. Theorem 5 separately requires  $\tau_N \geq C \log N$  and augmented-process mixing. Diurnal dependence motivates lag sensitivity.

**Lemma 1** (Exact orthogonality). *Let  $\psi_k(\theta; m, \mu) = \gamma_k(T - m) \{Y - \mu - \sum_j \gamma_j \theta_j (T - m)\}$ . At  $m = \mathbb{E}[T | F]$ ,  $\mu = \mathbb{E}[Y | F]$ , the Gateaux derivative of  $\mathbb{E}\psi_k$  in every direction  $(\delta_m, \delta_\mu)$  of square-integrable functions of  $F$  vanishes, for every  $\theta$  and every posterior  $\gamma$ ; so does that of  $\mathbb{E}[\gamma_j \gamma_k (T - m)^2]$ .*

*Proof.* Each derivative is an expectation of a product of an  $F$ -measurable factor (the weights  $\gamma$  and  $\delta$ ) and  $T - m$  or  $Y - \mu - \sum_j \gamma_j \theta_j (T - m)$ , whose conditional mean given  $F$  is zero.  $\square$

The natural alternative, used in earlier drafts of this work and closest to regime fixed effects, fits regime- $k$  nuisances on  $X$  with weights  $\gamma_{\cdot k}$ . Its Gram part is not orthogonal for soft weights: under a nuisance perturbation its score derivative is 85% of the Jacobian, against 0.4% for Lemma 1 (Appendix E). Section 4 shows why it fails, and we keep it as an ablation. Two simpler estimators on the same inputs are our main baselines: *hard regime fixed effects* (one-hot arg max weights) and *posterior-feature DML*, a single-index DML with nuisances on  $F_t$ . Algorithm 1 summarizes the procedure and the diagnostics we recommend reporting with it. Complexity is  $\mathcal{O}(B C_{\text{nuis}} + NK^2 + K^3)$ ; Appendix J checks  $K \leq 5$ .

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**Algorithm 1** OR-DML with a reliability report
 

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**Require:**  $\{Y_t, T_t, X_t, Z_t\}_{t=1}^N$ , folds  $B$ , penalty  $\lambda \geq 0$ .

- 1: Fit a Gaussian HMM to  $Z$  (multi-restart EM); compute posteriors  $\gamma_t$  (smoothing or causal filter).
  - 2: Embargo  $\tau^* = \max\{h : \max_j |\hat{\rho}_j(h)| \geq 1.96/\sqrt{N}\} + 1$ , at least  $C \log N$ .
  - 3: Cross-fit  $\hat{m}, \hat{\mu}$  on  $F_t = (X_t, \gamma_t \otimes X_t)$  over purged contiguous blocks.
  - 4: Form  $\hat{J}, \hat{S}$  by (3); return  $\hat{\theta}_\lambda = (\hat{J} + \lambda I)^{-1} \hat{S}$  with a HAC sandwich.
  - 5: **Report:** hard-FE and posterior-feature baselines;  $\hat{v}, \hat{a}, \widehat{\text{Var}}(\tilde{T})$  and the robustness value  $b^*$  (Section 4); raw and standardized  $\lambda_{\min}$ ; lead placebos aligned to the outcome's information set.
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## 4 HOW STATE UNCERTAINTY AND TASK CONDITIONING PROPAGATE

Let  $\mathbf{H}_t = \mathbf{e}_{S_t}$ ,  $\varepsilon_\gamma = \mathbb{E}\|\gamma_t - \mathbf{H}_t\|_1$ , and let  $\mathbf{J}, \mathbf{S}$  be the population versions of the coupled system at the population nuisances of either design. Define  $\boldsymbol{\theta}_\gamma = \mathbf{J}^{-1}\mathbf{S}$  and  $\mathbf{b}_\gamma = \mathbf{S} - \mathbf{J}\boldsymbol{\theta}^*$ , so  $\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^* = \mathbf{J}^{-1}\mathbf{b}_\gamma$  exactly.

**Theorem 3** (Graceful degradation). *Under Assumptions 2 and 4, suppose the nuisance targets are within  $C_\eta \varepsilon_\gamma$  of the oracle regime nuisances in  $L^1$ . This holds for posterior-weighted nuisances, and for posterior-adjusted nuisances when the posterior is calibrated given  $F$  and  $Z$  is independent of  $(T, Y)$  given  $(X, S)$ . If  $\lambda_{\min}(\mathbf{J}) > 0$ , then  $\|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 \leq C\varepsilon_\gamma/\lambda_{\min}(\mathbf{J})$  with explicit  $C$ , and  $\|\hat{\boldsymbol{\theta}}_\gamma - \boldsymbol{\theta}^*\|_2 \leq C\varepsilon_\gamma/\lambda_{\min}(\mathbf{J}) + \mathcal{O}_P((N^{-1/2} + \rho_N)/\lambda_{\min}(\mathbf{J}))$ , with  $\rho_N = r_N^2$  for posterior-adjusted nuisances (Lemma 1) and  $\rho_N = r_N$  for posterior-weighted ones, where  $r_N$  is the nuisance rate (Appendices D, E).*

The bound separates the two errors the abstract names: state recovery ( $\varepsilon_\gamma$ ) and nuisance estimation ( $\rho_N$ ). In a controlled design  $\varepsilon_\gamma/\lambda_{\min}(\hat{\mathbf{J}})$  ranks causal error almost perfectly (Spearman 0.995 over 4,900 runs, versus 0.626 for  $\varepsilon_\gamma$  and 0.521 for  $1/\lambda_{\min}$ ). The next two results explain why that success is fragile:  $\lambda_{\min}(\mathbf{J})$  is itself a function of the state error.

**The coupled score under state confusion.** Consider two regimes with stationary weights  $\pi_s$ , treatment means  $m_s$ , outcome means  $\mu_s = \theta_s m_s + g_s$  and innovation scales  $\sigma_s$ . Take regime-wise nuisances (the posterior-weighted design with correct regime means) and a proxy that places weight  $1 - \alpha$  on the true regime and  $\alpha = \varepsilon_\gamma/2$  on the other.

**Proposition 1** (Exact law for regime-weighted scores).

With  $\gamma_j(s) = 1 - \alpha$  if  $j = s$  and  $\alpha$  otherwise,

$$J_{jk} = \sum_s \pi_s \gamma_j(s) \gamma_k(s) [(m_s - m_j)(m_s - m_k) + \sigma_s^2],$$

$$S_j = \sum_s \pi_s \gamma_j(s) [(m_s - m_j)(\mu_s - \mu_j) + \theta_s \sigma_s^2].$$

In the symmetric case ( $\pi_s = \frac{1}{2}$ ,  $\sigma_s = \sigma$ ,  $\Delta = m_1 - m_0$ ,  $G = \mu_1 - \mu_0$ ),  $\mathbf{J}$  has eigenvectors  $(1, \pm 1)$  with eigenvalues  $\lambda_+ = \frac{1}{2}\sigma^2 + \frac{1}{2}\alpha^2\Delta^2$  and  $\lambda_- = \frac{1}{2}\sigma^2(1 - 2\alpha)^2 + \frac{1}{2}\alpha^2\Delta^2$ , and

$$\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^* = \frac{\alpha\Delta G - \frac{1}{2}\alpha^2\Delta^2(\theta_0 + \theta_1)}{2\lambda_+} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{(\theta_0 - \theta_1)(\alpha(1 - 2\alpha)\sigma^2 - \frac{1}{2}\alpha^2\Delta^2)}{2\lambda_-} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

**Corollary 1** (Phantom resolution and the error hump).

In the symmetric case: (a)  $\lambda_{\min}(\mathbf{J}) = \lambda_-$  exceeds its oracle value  $\sigma^2/2$  whenever  $\varepsilon_\gamma > 8\sigma^2/(\Delta^2 + 4\sigma^2)$ , so a confused state can look better conditioned than the true one; (b) when  $|\Delta G| \gg \sigma^2|\theta_0 - \theta_1|$ , the error is  $\approx \sqrt{2} \frac{\varepsilon_\gamma}{2} |\Delta G|/(\sigma^2 + \varepsilon_\gamma^2 \Delta^2/4)$ , maximized at  $\varepsilon_\gamma^* = 2\sigma/|\Delta|$  with peak  $|G|/(\sqrt{2}\sigma)$ , and decreasing in  $\varepsilon_\gamma$  beyond it; (c) the dominant displacement lies on the  $\lambda_+$  eigenvector, so the worst-direction bound of Theorem 3 is loose by  $\lambda_+/\lambda_-$ .

Proposition 1 is exact (verified symbolically in the supplement) and involves no sampling. Applied to the asymmetric factorial design it reproduces every cell's median error to within 0.98–1.02 $\times$  and locates the error peaks at  $\varepsilon_\gamma \in \{0.05, 0.10, 0.20\}$  for  $\sigma_1 \in \{0.08, 0.15, 0.30\}$ , exactly where they are observed (Figure 1). The factorial's near-perfect rank correlation is therefore an algebraic identity of the design, not a statistical regularity. It works partly because  $\lambda_{\min}(\hat{\mathbf{J}})$  inflates: replacing it by the design value  $\sigma_1^2$  lowers the cell-level correlation from 0.997 to 0.958. The leading displacement  $\alpha\Delta G$  is the product of state error with the regime contrasts that the weighted nuisances fail to remove.

**Posterior-adjusted designs.** Posterior-adjusted nuisances remove those contrasts up to what the posterior cannot explain. Let  $v_t = \gamma_t(1 - \gamma_t)$  be the posterior variance of the regime indicator, and let  $T_t = m(X_t) + aS_t + V_t$  and  $Y_t = \theta_{S_t}T_t + g(X_t) + bS_t + U_t$ .

**Theorem 4** (Residual-state bias law). *Let  $K = 2$  and suppose the posterior is calibrated given  $F$  ( $P(S_t=1 | F_t) = \gamma_t$ ),  $V_t$  is independent of  $(S, Z, X)$  with variance  $\sigma^2$ , nuisances are cross-fitted at rates with  $r_N^2 = o(N^{-1/2})$ , and the posterior forgets distant proxies (Appendix E). With score weights  $\mathbf{w}_t = (1 - \gamma_t, \gamma_t)$  (OR-DML) or  $\mathbf{w}_t \equiv 1$  (posterior-feature DML),*

$$\hat{\boldsymbol{\theta}} \rightarrow_p \bar{\boldsymbol{\theta}} + \mathbf{J}^{-1}\mathbf{b}_\gamma,$$

$$\mathbf{J} = \mathbb{E}[\mathbf{w}\mathbf{w}^\top(a^2v + \sigma^2)], \quad \mathbf{b}_\gamma = a\mathbb{E}[\mathbf{w}c v],$$

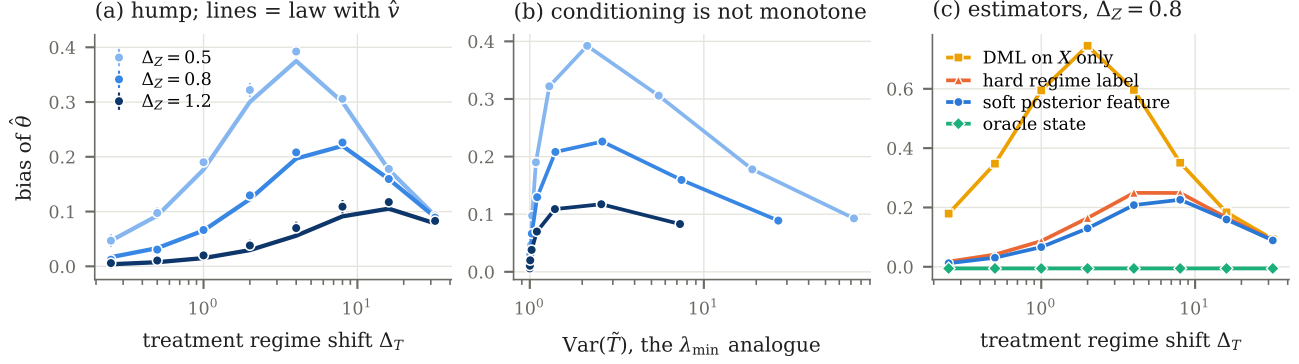


Figure 2: **The residual-state bias law with learned HMM posteriors** (pooled posterior-feature DML; smoother, linear nuisances, 12 replications per point, 95% bands). (a) Measured bias is hump-shaped in the treatment regime shift and matches the law evaluated with the observable plug-in  $\hat{v}$  (lines; across all 72 cells, correlation 0.996, median relative error 5.2%). (b) The same bias against  $\text{Var}(\tilde{T})$ , the scalar analogue of  $\lambda_{\min}(\hat{J})$ : conditioning improves monotonically while bias first rises and then falls. (c) Soft posterior features beat hard regime labels, as information monotonicity predicts; DML on  $X$  alone is far worse; the oracle state is unbiased.

with  $c_t = b + (\theta_1 - \theta_0)\{m(X_t) + a(1 - \gamma_t)\}$ ,  $\bar{\theta} = \theta^*$  for OR-DML and the  $(a^2v + \sigma^2)$ -weighted mean of  $\theta_0 + (\theta_1 - \theta_0)\gamma$  for the pooled estimator. Hence: (a) with no treatment regime shift ( $a = 0$ ) both are consistent for any calibrated posterior, whatever  $b$  and the effect heterogeneity; (b) for homogeneous effects the pooled bias is  $abv/(a^2v + \sigma^2)$ , hump-shaped in  $a$  with peak at  $|a| = \sigma/\sqrt{v}$  and  $\sup |B| = |b|\sqrt{v}/(2\sigma)$ , increasing in  $v$ , and a soft posterior feature is never worse than its hard label; (c)  $\hat{J}$  grows with  $a^2v$ , so phantom resolution is not specific to the weighted design.

The law is exact up to sampling error: over 9 configurations with  $N = 10^6$  the estimator and the law agree to within 0.008, and with  $a = 0$  the bias is 0.005 (Appendix E). The pooled form with  $K > 2$  regimes is  $a^\top \Sigma b / (a^\top \Sigma a + \sigma^2)$  with  $\Sigma$  the residual state covariance (Appendix G). Both laws have the same anatomy. A numerator proportional to state error times the product of the treatment and outcome regime shifts is divided by a residual treatment variance that the same treatment shift inflates. This is why  $\text{Var}(\tilde{T})$ , the scalar analogue of  $\lambda_{\min}(\hat{J})$ , cannot be read as a monotone certificate (Figure 2). The two designs differ in what multiplies the state error: weighted nuisances leak the raw contrast  $\Delta G$  at first order in  $\varepsilon_\gamma$ , adjusted nuisances only the residual variance  $v$ .

**Observability and inference.** For a calibrated posterior,  $\hat{v} = N^{-1} \sum_t \hat{\gamma}_t(1 - \hat{\gamma}_t)$  satisfies  $|\mathbb{E}\hat{v} - v| \leq \sqrt{CE} + CE$  in the  $L^2$  calibration error  $CE$ , and  $\text{Var}(\tilde{T})$  is estimable. If the calibrated regime-mean model identifies the treatment shift  $a$ , only the outcome shift  $b$  remains unidentified, so the robustness value  $b^* = |\hat{\theta}| \widehat{\text{Var}}(\tilde{T}) / \|\hat{\Sigma}\hat{a}\|$  (the smallest outcome shift that explains the estimate away) gives a one-parameter sensitivity analysis. For EM-fitted HMMs the plug-in

converges at rate  $N^{-1/2} + \sqrt{\delta_X}$ , where  $\delta_X$  is regime information carried by the controls (Proposition 3). Under misspecification,  $\hat{v}$  may understate or overstate residual state variance; sensitivity values then require calibration assumptions.

**Theorem 5** (Inference around the proxy target). *For posterior-adjusted nuisances, under Theorem 3 and the boundedness, augmented-process  $\beta$ -mixing, nuisance-rate and differentiability conditions in Appendix H, use  $\tau^* \geq C \log N$ ,  $C > 1/c_\beta$ , and a separated HMM calibration window. Then  $\sqrt{N}(\hat{\theta}_\lambda - \theta_\lambda) \rightarrow \mathcal{N}(0, \Sigma_\lambda^{\text{tot}})$ , including representation uncertainty. Here  $\theta_\lambda - \theta^* = (\hat{J} + \lambda I)^{-1}(\mathbf{b}_\gamma - \lambda \theta^*)$  and  $\|\mathbf{b}_\gamma\| \leq C\varepsilon_\gamma$ . Structural coverage requires displacement  $o(N^{-1/2})$ ;  $\sqrt{N}(C\varepsilon_\gamma + \lambda M_\theta)/(\lambda_{\min} + \lambda) \rightarrow 0$  suffices. Occupation weighting requires its joint influence covariance. Implemented HAC intervals condition on the fitted HMM; full-sample smoothing is outside this theorem.*

When proxy displacement is not negligible, structural Wald tests need not have nominal size. We therefore pair estimates with sensitivity analysis rather than an oracle bias correction.

## 5 EXPERIMENTS

**Learned posteriors break the scalar ratio.** To remove the synthetic-proxy circularity, we learn posteriors with a Gaussian HMM (EM, three restarts, smoothing) from noisy proxies across  $\Delta_Z \times \sigma_1 \times \Delta_T$  (36 cells, 720 runs,  $N = 1,200$ ). All diagnostics use the realized generated-state Gram matrix. Table 1 shows that the ratio  $\varepsilon_\gamma/\lambda_{\min}(\hat{J})$  falls from 0.995 to 0.517. Restoring the oracle geometry recovers 0.750. Restoring instead the numerator scale the laws predict, using the plug-in regime shifts  $\hat{\Delta}_T, \hat{\Delta}_Y$  of treat-

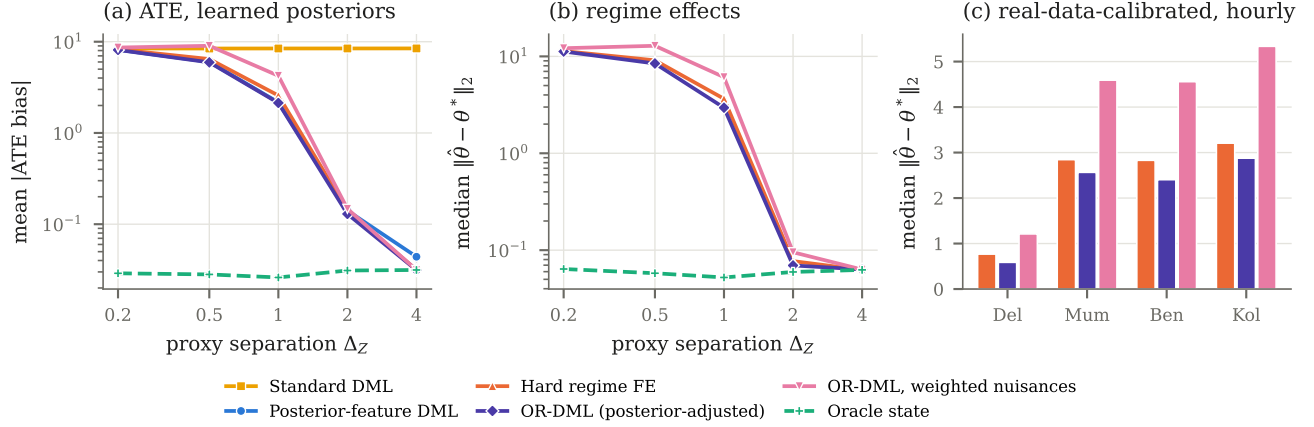


Figure 3: **Conditioning nuisances on the posterior, rather than weighting them by it, fixes the coupled estimator.** (a) Mean absolute ATE bias and (b) median regime-effect error along the observability frontier with learned HMM posteriors (100 datasets per point, persistence 0.88). Posterior-adjusted OR-DML is the most accurate regime-aware estimator at every  $\Delta_Z$  (ties at  $\Delta_Z = 4$ ); the posterior-weighted design is the least accurate wherever states are imperfectly recovered, as Proposition 1 predicts. Below  $\Delta_Z \approx 1$  every proxy-based estimator fails. (c) Regime-effect error in the real-data-calibrated semi-synthetic benchmark (30 datasets per city): the same ordering in every city.

Table 1: Spearman correlation between failure scores and causal error. Synthetic proxy: factorial design. Learned: EM posteriors; “within  $\Delta_T$ ” averages correlations at fixed treatment shift.

| Failure score                                                    | synthetic proxy |  | learned HMM posteriors |                         |
|------------------------------------------------------------------|-----------------|--|------------------------|-------------------------|
|                                                                  | runs            |  | runs                   | within $\Delta_T$ cells |
| $\varepsilon_\gamma$ (posterior error)                           | 0.626           |  | 0.187                  | 0.083 0.222             |
| $1/\lambda_{\min}(\hat{J})$                                      | 0.521           |  | 0.465                  | 0.483 0.509             |
| $\varepsilon_\gamma/\lambda_{\min}(\hat{J})$                     | 0.995           |  | 0.517                  | 0.388 0.564             |
| $\bar{H}/\lambda_{\min}(\hat{J})$ (observable)                   | —               |  | 0.581                  | 0.442 0.611             |
| $\varepsilon_\gamma/\lambda_{\min}(\hat{J}^*)$ (oracle geometry) | —               |  | 0.750                  | 0.727 0.783             |
| $\bar{H} \Delta_T\Delta_Y /\lambda_{\min}(\hat{J})$ (observable) | —               |  | 0.664                  | 0.730 0.675             |
| $\varepsilon_\gamma \Delta_T\Delta_Y /\lambda_{\min}(\hat{J})$   | —               |  | 0.695                  | 0.720 0.863             |

ment and outcome, gives 0.695 (0.863 across cells), and 0.664 with entropy in place of  $\varepsilon_\gamma$ , which is fully observable. With the same posteriors, hard fixed effects beat the posterior-weighted design in 100% of cells, while posterior-adjusted OR-DML beats hard fixed effects in 72% (median error ratio 0.97; 0.69 against the weighted design). All of its losses occur at the smallest treatment innovation, where the  $a^2v$  term dominates  $\mathbf{J}$ .

**Observability and persistence.** Table 2 (left) varies proxy separation  $\Delta_Z$  in the DGP of Theorem 1; a persistence sweep adds regime-0 stay probabilities 0.70 and 0.97 (900 datasets in all). Standard and blocked DML are stuck at the omitted-regime bias. All proxy-based estimators fail at  $\Delta_Z \leq 1$  and approach the oracle at  $\Delta_Z \geq 2$ . Among the 7 cells with  $\Delta_Z \geq 1$  (Figure 3), OR-DML is best or tied for the ATE in 7 and for regime effects in 7. At  $\Delta_Z = 2$  its ATE bias is 0.129 (hard fixed effects 0.141, weighted design 0.147) with coverage 91%, and at  $\Delta_Z = 1$  its regime-effect error is 2.954 against 3.652 and 6.107. Persistence helps

exactly through the posterior: at  $\Delta_Z = 1$ , raising the stay probability from 0.70 to 0.97 lowers  $\varepsilon_\gamma$  from 0.090 to 0.026 and OR-DML’s bias from 2.725 to 0.960 (hard fixed effects: 3.260 to 1.194). On the legacy version of this DGP the excess bias of soft conditioning over the oracle is predicted by the law to within 0.04 at every  $\Delta_Z$  (Appendix G.1), so the transition from failure to success is explained by  $v(\Delta_Z)$ , not by the spectrum of  $\hat{\mathbf{J}}$ . Under a continuous latent mixture that no two-state model describes, OR-DML beats hard fixed effects in 100% of 100 replications, while the weighted design beats them in 0% (Appendix L).

**Real-data-calibrated benchmark.** The right half of Table 2 keeps everything real except the causal effects. That includes meteorological proxies and controls, the observed  $\text{NO}_2$  series, the hourly missingness pattern, and regime persistence and overlap: true paths are drawn by forward-filtering backward-sampling from an HMM calibrated per city (stay probabilities 0.90–0.96). With posteriors that are exactly calibrated and 95–96% accurate, every proxy-based estimator retains large bias and zero coverage. For regime effects OR-DML is the most accurate regime-aware estimator in every city (Delhi 0.59, against 0.77 for hard fixed effects and 1.21 for the weighted design). For the pooled ATE it is best in Delhi and Kolkata, and posterior-feature DML is best in Mumbai and Bengaluru: scalar pooling avoids estimating weak regime contrasts. OR-DML’s advantage here concerns heterogeneous regime effects, not universal scalar-ATE dominance. The weighted design is worse than standard DML outside Delhi. The plug-in law, computed from the posterior, the fitted

Table 2: **Posterior-adjusted OR-DML is the most accurate regime-aware estimator, but no estimator removes residual-state bias.** *Left:* synthetic frontier with learned HMM posteriors (100 datasets per  $\Delta_Z$ ,  $N = 1,200$ , persistence 0.88, true ATE 1.53). *Right:* real-data-calibrated semi-synthetic benchmark on the corrected hourly grid (30 datasets per city,  $N = 2,324\text{--}9,034$ ; real meteorology,  $\text{NO}_2$  and gaps; true states drawn from the calibrated HMM’s smoothing distribution). Bold: best non-oracle within each block. Coverage, persistence variants and all estimators: Appendices L and M.

| Estimator                                                                | synthetic, proxy separation $\Delta_Z$ |             |             | real-data-calibrated, hourly |             |             |             |
|--------------------------------------------------------------------------|----------------------------------------|-------------|-------------|------------------------------|-------------|-------------|-------------|
|                                                                          | 1                                      | 2           | 4           | Del                          | Mum         | Ben         | Kol         |
| <i>ATE: mean absolute bias</i>                                           |                                        |             |             |                              |             |             |             |
| Oracle-state                                                             | 0.03                                   | 0.03        | 0.03        | 0.03                         | 0.07        | 0.12        | 0.14        |
| Standard DML                                                             | 8.43                                   | 8.44        | 8.44        | 1.94                         | 3.00        | 2.97        | 3.23        |
| Block DML                                                                | 8.47                                   | 8.47        | 8.48        | 2.16                         | 3.09        | 3.20        | 3.30        |
| DML + $Z$ controls                                                       | 5.43                                   | 2.63        | 0.86        | 1.22                         | 3.03        | 2.60        | 3.06        |
| Posterior-feature DML                                                    | 2.16                                   | 0.14        | 0.04        | 0.55                         | <b>1.03</b> | <b>1.51</b> | 2.17        |
| Hard regime FE                                                           | 2.57                                   | 0.14        | <b>0.03</b> | 0.55                         | 1.86        | 1.97        | 2.30        |
| OR-DML, weighted nuisances                                               | 4.24                                   | 0.15        | <b>0.03</b> | 0.88                         | 3.10        | 3.22        | 3.81        |
| <b>OR-DML</b> (posterior-adjusted)                                       | <b>2.13</b>                            | <b>0.13</b> | <b>0.03</b> | <b>0.42</b>                  | 1.65        | 1.69        | <b>2.05</b> |
| <i>Regime effects: median <math>\ \hat{\theta} - \theta^*\ _2</math></i> |                                        |             |             |                              |             |             |             |
| Hard regime FE                                                           | 3.65                                   | 0.08        | <b>0.06</b> | 0.77                         | 2.84        | 2.83        | 3.21        |
| OR-DML, weighted nuisances                                               | 6.11                                   | 0.10        | 0.06        | 1.21                         | 4.59        | 4.56        | 5.33        |
| <b>OR-DML</b> (posterior-adjusted)                                       | <b>2.95</b>                            | <b>0.07</b> | 0.06        | <b>0.59</b>                  | <b>2.56</b> | <b>2.40</b> | <b>2.88</b> |
| posterior error $\varepsilon_\gamma$                                     | 0.062                                  | <0.001      | <0.001      | 0.087                        | 0.097       | 0.088       | 0.083       |

treatment shift and the injected outcome shift, ranks the four cities’ biases correctly, with magnitudes within 7–37%. The generated-state  $\lambda_{\min}$  exceeds its oracle value by 1.7–2.4 $\times$  in Mumbai, Bengaluru and Kolkata (1.04 $\times$  in Delhi).

**Representation quality is not task reliability.** Across 15 shifted worlds and four encoders (HMM, GRU, Transformer, linear SSM), adding posterior NLL to state-recovery  $F_1$  raises leave-one-world-out prediction of downstream reliability from  $R^2 = 0.113$  to 0.221. Yet temperature scaling cuts GRU/Transformer ECE by 47.9%/60.3% without improving failure discrimination ( $\Delta\text{AUC} -0.0035/-0.0075$ ). By Theorem 4, recalibration changes the *estimate*  $\hat{v}$  but not the residual information the downstream task actually has (Appendix N).

## 6 OBSERVATIONAL STRESS TEST: FOUR CITIES

We apply OR-DML and the sensitivity analysis to  $\text{NO}_2 \rightarrow \text{PM}_{2.5}$  in Delhi, Mumbai, Bengaluru and Kolkata (OpenAQ/CPCB stations and Open-Meteo weather, Feb 2025–Jun 2026) on a *corrected* hourly grid. An audit found that the legacy panel rounded half-hour timestamps with ties-to-even, collapsing it onto a two-hour grid, so “hourly” lags were two hours. This changed Delhi’s single-index posterior-feature estimate from 0.58 to 0.37; Table 3 instead reports the posterior-occupation-weighted coupled estimate 0.234 (Appendix O). Both are exposure associations, not

known true ATEs. Placebo leads are aligned to the outcome’s own information set. Without this alignment the one-hour lead is mechanically zero, because the lagged outcome is a control. Real posteriors are sharp here, so the weighted design gives the same ATE to within 0.12 outside Kolkata (Delhi 0.241).

Table 3 separates four axes the reliability rule conflates. *Scale:* raw  $\lambda_{\min}$  ranges from 0.0012 to 63.2, but the standardized value only from 0.20 to 0.46. Raw conditioning reflects units and sensor resolution, not collinearity. *Data quality:* Kolkata’s  $\text{NO}_2$  takes 16 distinct values, and Mumbai’s longest constant run is 277 h; screening all qualifying runs moves the ATE from 3.02 to 0.38 ( $N$  8,113 $\rightarrow$ 6,256).<sup>1</sup> *Temporal validity:* Bengaluru and screened Kolkata fail lead placebos, the latter strongly ( $p = 0.0003$ ), so the apparently significant screened Kolkata effect (3.32) is not interpretable. *Residual confounding:* Delhi’s association is positive in both regimes (0.430 and 0.102; ATE 0.234) and would need an outcome shift of 82 SD (90% interval 54–144) to be explained away through the HMM-regime channel. Its one-hour lead placebo nonetheless rejects (0.072,  $p = 0.012$ , about a third of the contemporaneous coefficient). This points to short-horizon confounding

<sup>1</sup>Mumbai loses 1,857 complete cases (22.9%), across all flagged runs rather than one episode. This sensitivity exclusion does not establish hardware failure or validate the remaining sensor calibration. Five EM restarts also span 1,182 log-likelihood units (Appendix G); neither optimizer multimodality nor misspecification establishes the direction of error in  $\hat{v}$  or  $b^*$ .

outside that channel. We therefore report Delhi as an exposure association that is robust to residual *regime* confounding, not as an emissions effect, and the other three cities as inconclusive.

## 7 DISCUSSION

The intuition we set out to test was that a confident latent state and a well-conditioned score Jacobian together certify a causal estimate. They do not. A generated state enters the downstream estimating equation twice, in its bias and in its apparent information, and the two move together. The scalar  $\varepsilon_\gamma/\lambda_{\min}$  is an honest worst-case bound and an excellent ranking device when the state error is synthetic and the regime shifts are fixed. It degrades sharply once posteriors are learned, because its denominator is partly made of the error it is meant to measure. Two rules survive. *Design*: condition the nuisances on the posterior and use it only to weight the target. This makes the coupled score exactly orthogonal and gave the most accurate regime estimates in every benchmark here. *Reporting*: state the residual state variance, the regime shifts it multiplies and a robustness value; compare against hard fixed effects and posterior-feature baselines; and test temporal validity with aligned placebos.

**Outlook.** Nothing in the mechanism is specific to HMMs. Any generated state (a neural encoder’s regime embedding, a clustering label, a changepoint segmentation) enters a downstream estimating equation both as an adjustment and as a source of apparent information, and the same accounting applies: what matters is the residual state variance the representation leaves, multiplied by the regime shifts of treatment and outcome, not the representation’s accuracy or the conditioning of the score it feeds. Evaluating representations for causal use therefore means reporting that residual, and designing estimators so that the representation is used as an orthogonal adjustment rather than as a weight on the nuisance fit.

**Limitations.** The laws are probability limits under linear regime shifts, exogenous treatment innovations and calibrated posteriors. The robustness value bounds rather than corrects, and depends on calibration under misspecified emissions. Posterior-adjusted OR-DML does not remove residual-state bias: under weak observability every proxy-based estimator, ours included, has large bias and no coverage, and for the pooled ATE posterior-feature DML is sometimes more accurate. The city analysis is observational. Proxies overlap controls, lagged pollutants may be mediators, and Delhi’s lead placebo rejects.

Table 3: Four-city stress test on the corrected hourly grid (OR-DML, HAC lag 24 h).  $\lambda_{\min}$  is raw,  $\lambda_{\min}^{\text{std}}$  is scale-free. Lead placebo: smallest  $p$ -value over leads  $h \in \{-6, -3, -1\}$  h.  $b^*$ : robustness value in SD of  $\tilde{Y}$  (posterior-feature analysis; for screened rows, its own screened sample). Screened: runs of  $\geq 6$  identical  $\text{NO}_2$  readings removed (no change for Delhi).

| City                 | $N$   | $\lambda_{\min}$ | $\lambda_{\min}^{\text{std}}$ | $\hat{\theta}_{\text{ATE}}$ (SE) | $\hat{\theta}_1$ (SE) | $\hat{\theta}_2$ (SE) | lead placebo                       | $b^*$ (SD) |
|----------------------|-------|------------------|-------------------------------|----------------------------------|-----------------------|-----------------------|------------------------------------|------------|
| Delhi                | 9,024 | 63.2             | 0.46                          | 0.234 (0.029)                    | 0.430 (0.053)         | 0.102 (0.033)         | rejects ( $h = -1$ , $p = 0.012$ ) | 82.3       |
| Mumbai               | 8,113 | 0.0851           | 0.41                          | 3.017 (1.682)                    | 5.480 (2.416)         | 1.189 (2.429)         | passes ( $p_{\min} = 0.33$ )       | 11.9       |
| Bengaluru            | 7,775 | 2.4              | 0.39                          | 0.004 (0.074)                    | 0.131 (0.140)         | -0.132 (0.053)        | rejects ( $h = -6$ , $p = 0.016$ ) | 26.8       |
| Kolkata              | 2,322 | 0.00116          | 0.40                          | 1.217 (1.143)                    | -0.674 (0.455)        | 3.894 (2.808)         | passes ( $p_{\min} = 0.10$ )       | 10.9       |
| Mumbai (screened)    | 6,256 | 0.0495           | 0.20                          | 0.383 (1.945)                    | -4.012 (4.687)        | 2.757 (1.496)         | rejects ( $h = -6$ , $p = 0.047$ ) | 1.2        |
| Bengaluru (screened) | 7,180 | 2.3              | 0.40                          | 0.020 (0.082)                    | 0.185 (0.155)         | -0.156 (0.062)        | rejects ( $h = -3$ , $p = 0.015$ ) | 35.3       |
| Kolkata (screened)   | 1,422 | 0.00172          | 0.40                          | 3.319 (1.397)                    | 0.191 (0.541)         | 6.916 (2.981)         | rejects ( $h = -1$ , $p = 0.000$ ) | 16.5       |

## AI Use Statement

Large language models assisted with code, derivations, experiment scripting and drafting. Every number in this paper is generated from committed artifacts by a single script (`src/synthesis/make_tables_numbers.py`) and checked by `verify_submission.py`, which re-derives all tables and macros and fails on any mismatch. The closed forms are verified symbolically, and earlier drafts' errors found in audit (time grid, placebo alignment, overstated estimator claims) are documented in Appendix Q. The authors reviewed all arguments and take full responsibility for the content.

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## CHECKLIST

1. For all models and algorithms presented, check if you include:
  - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. **[Yes]** Sections 2–4, Appendix A, Algorithm 1.
  - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. **[Yes]** Lemma 1, Theorems 3, 4, 5, 6; complexity and scalability in Appendix J.
  - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. **[Yes]** Supplementary archive with `requirements.txt`, `reproduce.sh` and `verify_submission.py`.
2. For any theoretical claim, check if you include:
  - (a) Statements of the full set of assumptions of all theoretical results. **[Yes]** Appendix A; Assumptions 5–7, 8, 9; setting (E) for learned HMMs.
  - (b) Complete proofs of all theoretical results. **[Yes]** Appendices B–I. The change-point scaling of the residual variance is summarized here and proved in the supplementary archive; it is not used for any main-text claim.
  - (c) Clear explanations of any assumptions. **[Yes]** Including where they fail on the real data ( $\delta_X > 0$ ; multimodal likelihood for Mumbai; exclusion restrictions for proximal identification).
3. For all figures and tables that present empirical results, check if you include:
  - (a) The code, data, and instructions needed to reproduce the main experimental results. **[Yes]** Appendix P; raw sensor files are included.
  - (b) All the training details (e.g., data splits, hyperparameters, how they were selected). **[Yes]** Appendices K–O.
  - (c) A clear definition of the specific measure or statistics and error bars. **[Yes]** Mean absolute bias, RMSE and coverage over stated replications; HAC standard errors with stated lags; block-bootstrap intervals for robustness values.
  - (d) A description of the computing infrastructure used. **[Yes]** All experiments run on a two-core CPU machine; runtimes in Appendix P.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets:
  - (a) Citations of the creator if your work uses existing assets. **[Yes]** OpenAQ/CPCB and Open-Meteo.
  - (b) The license information of the assets, if applicable. **[Yes]** Public air-quality and reanalysis data under their providers’ open-data terms.
  - (c) New assets either in the supplemental material or as a URL, if applicable. **[Yes]** Code, processed data and all result files in the supplementary archive.
  - (d) Information about consent from data providers/curators. **[Not Applicable]** Aggregate environmental measurements.
  - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. **[Not Applicable]**
5. If you used crowdsourcing or conducted research with human subjects:
  - (a) The full text of instructions given to participants and screenshots. **[Not Applicable]**
  - (b) Descriptions of potential participant risks, with links to IRB approvals if applicable. **[Not Applicable]**
  - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. **[Not Applicable]**

## SUPPLEMENTARY MATERIAL

**Roadmap.** Appendix A states the assumptions. Appendices B–D prove Theorems 1–3. Appendix E treats posterior-adjusted nuisances: Lemma 1, Theorem 3 for that design, the proof of Theorem 4 and its numerical checks. Appendix F proves Proposition 1 and Corollary 1, and records the phantom-resolution results for hard classifiers and higher dimensions. Appendix G gives the pooled law for  $K \geq 2$  regimes and its learned-HMM extension. Appendix H covers inference. Appendix I gives the operational surrogate, proxy screening and spectral regularization. Appendix J gives the algorithm and scalability. Appendices K–N document every experiment. Appendix O documents the sensor data and its audit. Appendix P explains how to reproduce and verify every number. Appendix Q records what changed relative to earlier drafts and why.

## A ASSUMPTIONS

**Assumption 1** (Regime-Conditional Unconfoundedness & Innovation Exogeneity). Conditional on observed covariates  $X_t$ , latent regime  $S_t$ , and proxy history  $Z_{1:N}$ , potential outcomes are unconfounded ( $Y_t(u) \perp T_t \mid (X_t, S_t, Z_{1:N}), \forall u$ ) and structural innovations have zero mean:

$$\mathbb{E}[U_t \mid X_t, S_t, T_t, Z_{1:N}] = 0. \quad (4)$$

**Assumption 2** (Overlap, Envelopes & State Positivity). There exist constants  $\eta, c_0 > 0$ ,  $B_T, B_Y < \infty$ , and  $c_1, c_2 > 0$  such that:

1. **State Positivity & Envelopes:** Stationary distribution satisfies  $\min_k \pi_k^* \geq \eta$ , and for almost every  $x$  and all  $k$ ,  $P(S_t = k \mid X_t = x) \geq c_0$ ,  $\mathbb{E}[\gamma_{tk} \mid X_t = x] \geq c_0$ . Variables satisfy  $|T_t| \leq B_T$ ,  $|Y_t| \leq B_Y$  a.s., so  $|\tilde{T}_{tk}| \leq M_T \equiv 2B_T$ ,  $|Y_{tk}| \leq M_Y \equiv 2B_Y$ .
2. **Within-Regime Treatment Overlap:** For all  $k \in \{1, \dots, K\}$  and almost everywhere on  $\text{supp}(X_t)$ :

$$c_1 \leq \text{Var}(T_t \mid X_t, S_t = k) \leq c_2. \quad (5)$$

**Assumption 3** (Latent Proxy Sufficiency). The auxiliary state proxy sequence  $Z_{1:N}$  satisfies latent proxy sufficiency:  $P(S_t = k \mid X_t, T_t, Z_{1:N}) = P(S_t = k \mid Z_{1:N}) \equiv \gamma_{tk}$  (or  $P(S_t = k \mid Z_{1:t})$  for causal forward filtering) almost surely for all  $k \in \{1, \dots, K\}$ . *Discussion.* That is,  $Z_{1:N}$  screens off  $S_t$  from  $(X_t, T_t)$ . Combined with Assumption 1, this guarantees  $\mathbb{E}[Y_t \mid X_t, T_t, Z_{1:N}] = \sum_{k=1}^K \gamma_{tk} [\theta_k^* T_t + g_k(X_t)]$ . Notably, our simulation DGPs deliberately violate Assumption 3 ( $S_t \not\perp T_t \mid Z_t$ ) via regime-dependent treatment means; crucially, Theorem 3 does not require Assumption 3, depending only on envelopes and rates (Assumptions 2 and 4).

**Assumption 4** (Nuisance Estimation and Mixing Rates). The data process  $(Y_t, T_t, X_t, Z_t)$  is strictly stationary and geometrically  $\alpha$ -mixing with  $\alpha(m) \leq C_\alpha \exp(-c_\alpha m)$  for  $C_\alpha, c_\alpha > 0$ . Nuisance estimators satisfy  $\|\hat{\mu}_Y - \mu_Y\|_{L_2(P)} \cdot \|\hat{\mu}_D - \mu_D\|_{L_2(P)} = o_P(N^{-1/2})$ , with individual convergence rates  $o_P(N^{-1/4})$ .

## B PROOF OF THEOREM 1

**Full statement.** Under (1)–(2), with  $\tilde{T}_t = T_t - \mathbb{E}[T_t \mid X_t]$ ,

$$\mathcal{B}_{\text{het}} = \frac{\sum_k \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (6)$$

$$\mathcal{B}_{\text{conf}} = \frac{\sum_k \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (7)$$

with  $\pi_k(X) = P(S_t = k \mid X_t)$ ,  $\bar{m}(X) = \mathbb{E}[T_t \mid X_t]$ ,  $\Delta m_k(X) = \mathbb{E}[T_t \mid X_t, S_t = k] - \bar{m}(X)$  and  $\Delta g_k(X) = g(X, k) - \mathbb{E}[g(X_t, S_t) \mid X_t]$ . Consider the Structural Causal Model (1)–(2):

$$\begin{aligned} T_t &= m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t \mid X_t, S_t] = 0, \\ Y_t &= \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t \mid X_t, S_t, T_t] = 0. \end{aligned}$$

Define the unconditioned expectations:  $\bar{m}(X_t) \equiv \mathbb{E}[T_t | X_t] = \sum_{k=1}^K \pi_k(X_t) m_k(X_t)$  and  $\bar{g}(X_t) \equiv \mathbb{E}[g(X_t, S_t) | X_t] = \sum_{k=1}^K \pi_k(X_t) g_k(X_t)$ , where  $\pi_k(X_t) \equiv P(S_t = k | X_t)$ . Residuals from cross-fitting on  $X_t$  alone satisfy  $\tilde{T}_t = T_t - \bar{m}(X_t) = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$ , where  $\Delta m_k(X_t) = m_k(X_t) - \bar{m}(X_t)$ . The outcome residual is:

$$\tilde{Y}_t = Y_t - \mathbb{E}[Y_t | X_t] = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) + g(X_t, S_t) - \bar{g}(X_t) + U_t. \quad (8)$$

Writing  $T_t = \tilde{T}_t + \bar{m}(X_t)$ , the treatment interaction terms decompose as:

$$\sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)]. \quad (9)$$

Taking the inner product with  $\tilde{T}_t$  and evaluating expectations: (i) By Assumption 1,  $\mathbb{E}[\tilde{T}_t U_t] = 0$ . (ii)  $\mathbb{E}[\tilde{T}_t \sum_k \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t] = \sum_{k=1}^K \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2] = \mathbb{E}[\tilde{T}_t^2] \sum_{k=1}^K \omega_k \theta_k^*$ , where  $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$  with  $\sum_k \omega_k = 1$ . (iii) Since  $\mathbb{E}[\tilde{T}_t | X_t] = 0$  and  $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) | X_t] = \Delta m_k(X_t) \pi_k(X_t)$ , the mean adjustment yields  $\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]$ . (iv) The confounding component evaluates to  $\mathbb{E}[\tilde{T}_t (g(X_t, S_t) - \bar{g}(X_t))] = \sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]$ . Dividing by  $\mathbb{E}[\tilde{T}_t^2]$  yields (6) and (7). ■

## C PROOF OF THEOREM 2

We exhibit two models satisfying Assumptions 1–3 with the same observable law for every sequence length. Let  $S_t$  be independent Bernoulli(1/2), so both rows of the transition matrix are (1/2, 1/2). Let  $Z_t$  be independent standard Gaussian noise,  $X_t$  independent of  $(S_t, Z_t)$ , and  $T_t$  independent Bernoulli(1/2). All sequences are independent across time. Thus

$$P(S_t = s | X_t, T_t, Z_{1:N}) = P(S_t = s | Z_{1:N}) = 1/2,$$

which verifies proxy sufficiency rather than contradicting it. State positivity holds with  $c_0 = \eta = 1/2$ , treatment variance is 1/4 in both states, and all treatment and outcome variables are bounded.

Generate independent uniforms  $E_t$  and define potential outcomes  $Y_t(a) = \mathbb{I}\{E_t \leq g_{S_t} + \theta_{S_t} a\}$  for  $a \in \{0, 1\}$ . Set  $Y_t = Y_t(T_t)$  and  $U_t = Y_t - g_{S_t} - \theta_{S_t} T_t$ . The probabilities below lie in  $[0, 1]$ ; treatment is independent of potential outcomes, and  $\mathbb{E}[U_t | X_t, S_t, T_t, Z_{1:N}] = 0$ . Also  $V_t = T_t - 1/2$  has conditional mean zero.

| Model | $\theta_0$ | $\theta_1$ | $g_0$ | $g_1$ |
|-------|------------|------------|-------|-------|
| $P_1$ | 0.20       | 0.20       | 0.30  | 0.50  |
| $P_2$ | 0.10       | 0.30       | 0.40  | 0.40  |

In both models,  $P(Y_t = 1 | T_t = 0) = 0.40$  and  $P(Y_t = 1 | T_t = 1) = 0.60$ . Since  $Y_t$  is binary, these specify its entire conditional distribution. The distribution of  $(T, X, Z)$  is also identical, and temporal independence gives equality of the observable sequence laws for every  $N \geq 1$ . The effect vectors (0.20, 0.20) and (0.10, 0.30) are not permutations. The enumeration in `src/synthesis/nonidentification.check.py` verifies the probability tables, posterior sufficiency and overlap.

**Scope.** This proves non-identification of the *regime-specific vector* in the stated memoryless model class. Both models have population ATE 0.20, which randomized treatment identifies; the example does not establish ATE non-identification. It does not imply that every persistent latent model with uninformative  $Z$  is unidentified: outcomes and treatment histories can supply identifying information under additional joint-model restrictions (Allman et al., 2009). □

## D PROOF OF THEOREM 3

This appendix proves Theorem 3 for posterior-weighted nuisances; Appendix E adapts the argument to posterior-adjusted nuisances. **Full statement.** Let  $\mathbf{H}_t = \mathbf{e}_{S_t}$ ,  $\gamma_t \in \Delta^K$  a state proxy with  $\varepsilon_\gamma \equiv \mathbb{E} \|\gamma_t - \mathbf{H}_t\|_1$ , and  $\mathbf{J}, \mathbf{S}$  the population coupled Gram matrix and score at the proxy nuisances  $(m_k^\gamma, \mu_k^\gamma)$ . Under Assumption 2

and  $\|\theta^*\|_2 \leq M_\theta$ , if  $\lambda_{\min}(\mathbf{J}) > 0$ : (i)  $\theta_\gamma \equiv \mathbf{J}^{-1}\mathbf{S}$  satisfies  $\|\theta_\gamma - \theta^*\|_2 \leq C\varepsilon_\gamma/\lambda_{\min}(\mathbf{J})$  with  $C = C_S + M_\theta C_J$ ,  $C_S = \sqrt{K}(M_T M_Y + M_Y \tilde{C}_m + M_T \tilde{C}_\mu)$ ,  $C_J = K(2M_T^2 + 2M_T \tilde{C}_m)$ ,  $\tilde{C}_m = 2B_T/c_0^2$  and  $\tilde{C}_\mu = 2B_Y/c_0^2$ . (ii) With  $r_N = \max_k(\|\hat{m}_k - m_k^\gamma\|_{L_2} + \|\hat{\mu}_k - \mu_k^\gamma\|_{L_2})$ , under purged cross-fitting and Assumption 4, on  $\mathcal{E}_N = \{\|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}} \leq \lambda_{\min}(\mathbf{J})/2\}$ ,  $\|\hat{\theta}_\gamma - \theta^*\|_2 \leq C\varepsilon_\gamma/\lambda_{\min}(\mathbf{J}) + \mathcal{O}_P((N^{-1/2} + r_N)/\lambda_{\min}(\mathbf{J}))$ .

Let  $\mathbf{H}_t = \mathbf{e}_{S_t} \in \{0, 1\}^K$  denote the unobserved true regime indicator vector. Under SCM (1)–(2), the oracle regime nuisance functions are:

$$m_k^*(X_t) \equiv \mathbb{E}[T_t | X_t, S_t = k], \quad \mu_k^*(X_t) \equiv \mathbb{E}[Y_t | X_t, S_t = k] = \theta_k^* m_k^*(X_t) + g(X_t, k). \quad (10)$$

**Step 1: Define proxy-weighted targets and population estimand.** The empirical cross-fitted learners  $\hat{m}_k$  and  $\hat{\mu}_k$  minimize the posterior-weighted empirical risk targeting the proxy-weighted conditional expectations:

$$m_k^\gamma(X_t) \equiv \frac{\mathbb{E}[\gamma_{tk} T_t | X_t]}{\mathbb{E}[\gamma_{tk} | X_t]}, \quad \mu_k^\gamma(X_t) \equiv \frac{\mathbb{E}[\gamma_{tk} Y_t | X_t]}{\mathbb{E}[\gamma_{tk} | X_t]}. \quad (11)$$

Define the centered residuals at proxy targets:  $\tilde{T}_{tk}^\gamma \equiv T_t - m_k^\gamma(X_t)$ ,  $\tilde{Y}_{tk}^\gamma \equiv Y_t - \mu_k^\gamma(X_t)$ , and at oracle targets:  $\tilde{T}_{tk}^* \equiv T_t - m_k^*(X_t)$ ,  $\tilde{Y}_{tk}^* \equiv Y_t - \mu_k^*(X_t)$ . The population proxy Gram matrix  $\mathbf{J} \in \mathbb{R}^{K \times K}$  and score vector  $\mathbf{S} \in \mathbb{R}^K$  are:

$$J_{jk} \equiv \mathbb{E}[\gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma], \quad S_k \equiv \mathbb{E}[\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma], \quad (12)$$

and their oracle counterparts  $\mathbf{J}^*, \mathbf{S}^*$  are defined by:

$$J_{jk}^* \equiv \mathbb{E}[H_{tj} H_{tk} \tilde{T}_{tj}^* \tilde{T}_{tk}^*] = \delta_{jk} \mathbb{E}[H_{tk} (\tilde{T}_{tk}^*)^2], \quad S_k^* \equiv \mathbb{E}[H_{tk} \tilde{T}_{tk}^* \tilde{Y}_{tk}^*], \quad (13)$$

where  $H_{tj} H_{tk} = \delta_{jk} H_{tk}$  because latent regimes are mutually exclusive.

**Step 2: Orthogonality structure of the coupled score (Lemma 2).**

**Lemma 2** (Orthogonality Structure of the Coupled Score). *Let  $\Psi_k(\theta; \eta) = \gamma_k \tilde{T}_k (\tilde{Y}_k - \sum_j \gamma_j \tilde{T}_j \theta_j)$  with  $\tilde{T}_k = T - m_k(X)$ ,  $\tilde{Y}_k = Y - \mu_k(X)$ , so  $\hat{\mathbf{S}} - \hat{\mathbf{J}}\theta = \frac{1}{N} \sum_t \Psi_t$ . At  $\eta^\gamma = (m^\gamma, \mu^\gamma)$  and any  $\theta$ , with  $g(X) \equiv \mathbb{E}[\|\gamma - \mathbf{H}\|_1 | X]$ :*

- (i)  $\partial_\mu \mathbb{E} \Psi_k[\Delta \mu] = 0$  for any proxy  $\gamma$ .
- (ii) The cross-moment part  $\gamma_k \tilde{T}_k \tilde{Y}_k$  is orthogonal in both directions.
- (iii)  $|\partial_m \mathbb{E} \Psi_k[\Delta m]| \leq 2M_T \|\theta\|_1 \sqrt{2\varepsilon_\gamma} \max_i \|\Delta m_i\|_{L_2}$ . For hard assignments  $\gamma_t \in \{0, 1\}^K$ , this derivative is identically 0.

*Proof.* (i)–(ii):  $\mathbb{E}[\gamma_k (T - m_k^\gamma) | X] = 0$  and  $\mathbb{E}[\gamma_k (Y - \mu_k^\gamma) | X] = 0$  by definition of  $m_k^\gamma, \mu_k^\gamma$ . (iii): differentiating  $-\gamma_k \tilde{T}_k \sum_j \gamma_j \tilde{T}_j \theta_j$  in direction  $\Delta m$  gives  $\sum_j \theta_j \mathbb{E}[\Delta m_k \gamma_k \gamma_j \tilde{T}_j + \Delta m_j \gamma_k \gamma_j \tilde{T}_k]$ . For  $j = k$ ,  $\mathbb{E}[\gamma_k^2 \tilde{T}_k | X] = \mathbb{E}[(\gamma_k^2 - \gamma_k) \tilde{T}_k | X]$  since  $\mathbb{E}[\gamma_k \tilde{T}_k | X] = 0$ , bounded by  $M_T |\gamma_k - H_k|$ . For  $j \neq k$ ,  $\gamma_k \gamma_j \leq |\gamma_k - H_k| + |\gamma_j - H_j|$ . Bounding each moment by  $M_T g(X)$  and using  $\|g\|_{L_2} \leq \sqrt{2\varepsilon_\gamma}$  yields (iii). For hard assignments,  $\gamma_k \gamma_j = \delta_{kj} \gamma_k$  so both moments vanish.  $\square$

**Step 3: Isolate proxy-induced bias via oracle score identity.** Under SCM (1)–(2), on the event  $S_t = k$  ( $H_{tk} = 1$ ), the oracle outcome residual is  $\tilde{Y}_{tk}^* = Y_t - \mu_k^*(X_t) = \theta_k^* \tilde{T}_{tk}^* + U_t$ . Substituting into  $S_k^*$ :

$$S_k^* = \mathbb{E} \left[ H_{tk} \tilde{T}_{tk}^* (\theta_k^* \tilde{T}_{tk}^* + U_t) \right] = \theta_k^* \mathbb{E}[H_{tk} (\tilde{T}_{tk}^*)^2] + \mathbb{E} \left[ H_{tk} \tilde{T}_{tk}^* \mathbb{E}[U_t | X_t, S_t, T_t, Z_{1:N}] \right] = \theta_k^* J_{kk}^*, \quad (14)$$

where the innovation term vanishes by Assumption 1. Because  $\mathbf{J}^* = \text{diag}(J_{11}^*, \dots, J_{KK}^*)$ , this yields the exact vector identity:

$$\mathbf{S}^* = \mathbf{J}^* \theta^*. \quad (15)$$

Now consider the proxy score discrepancy  $\mathbf{b}_\gamma \equiv \mathbf{S} - \mathbf{J}\theta^*$ . Expanding around the oracle identity:

$$\mathbf{b}_\gamma = (\mathbf{S} - \mathbf{S}^*) - (\mathbf{J} - \mathbf{J}^*)\theta^*. \quad (16)$$

**Step 4: Bound score and Gram proxy discrepancies.** We bound each component via telescoping expansions in  $\varepsilon_\gamma \equiv \mathbb{E}[\|\gamma_t - \mathbf{H}_t\|_1]$ . First, under positivity (Assumption 2),  $\pi_k(x) \equiv \mathbb{E}[H_{tk} | X_t = x] \geq c_0 > 0$  and  $\pi_k^\gamma(x) \equiv \mathbb{E}[\gamma_{tk} | X_t = x] \geq c_0 > 0$ . For any bounded variable  $V_t$  with  $|V_t| \leq M_V$  almost surely:

$$\left| \frac{\mathbb{E}[\gamma_{tk} V_t | X_t]}{\pi_k^\gamma(X_t)} - \frac{\mathbb{E}[H_{tk} V_t | X_t]}{\pi_k(X_t)} \right| \leq \frac{\mathbb{E}[|\gamma_{tk} - H_{tk}| |V_t| | X_t]}{\pi_k^\gamma(X_t)} + \mathbb{E}[H_{tk} |V_t| | X_t] \left| \frac{\pi_k(X_t) - \pi_k^\gamma(X_t)}{\pi_k^\gamma(X_t) \pi_k(X_t)} \right|$$

$$\leq \frac{2M_V}{c_0^2} \mathbb{E}[|\gamma_{tk} - H_{tk}| \mid X_t]. \quad (17)$$

Applying this bound to  $V_t = T_t$  and  $V_t = Y_t$  yields  $\mathbb{E}[m_k^\gamma(X_t) - m_k^*(X_t)] \leq \tilde{C}_m \varepsilon_\gamma$  and  $\mathbb{E}[\mu_k^\gamma(X_t) - \mu_k^*(X_t)] \leq \tilde{C}_\mu \varepsilon_\gamma$ . Second, expanding the score difference  $S_k - S_k^* = \mathbb{E}[\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma - H_{tk} \tilde{T}_{tk}^* \tilde{Y}_{tk}^*]$ :

$$\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma - H_{tk} \tilde{T}_{tk}^* \tilde{Y}_{tk}^* = (\gamma_{tk} - H_{tk}) \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma - H_{tk} (m_k^\gamma - m_k^*) \tilde{Y}_{tk}^\gamma - H_{tk} \tilde{T}_{tk}^* (\mu_k^\gamma - \mu_k^*). \quad (18)$$

Under bounded support ( $\sup_t |\tilde{T}_{tk}^\gamma| \leq M_T$ ,  $\sup_t |\tilde{Y}_{tk}^\gamma| \leq M_Y$ ), taking expectations yields  $|S_k - S_k^*| \leq (M_T M_Y + M_Y \tilde{C}_m + M_T \tilde{C}_\mu) \varepsilon_\gamma \equiv C_S \varepsilon_\gamma$ . Third, expanding the Gram difference  $J_{jk} - J_{jk}^* = \mathbb{E}[\gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma - H_{tj} H_{tk} \tilde{T}_{tj}^* \tilde{T}_{tk}^*]$ :

$$\begin{aligned} \gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma - H_{tj} H_{tk} \tilde{T}_{tj}^* \tilde{T}_{tk}^* &= (\gamma_{tj} \gamma_{tk} - H_{tj} H_{tk}) \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma - H_{tj} H_{tk} (m_j^\gamma - m_j^*) \tilde{T}_{tk}^\gamma \\ &\quad - H_{tj} H_{tk} \tilde{T}_{tj}^* (m_k^\gamma - m_k^*). \end{aligned} \quad (19)$$

Since  $|\gamma_{tj} \gamma_{tk} - H_{tj} H_{tk}| \leq |\gamma_{tj} - H_{tj}| + |\gamma_{tk} - H_{tk}|$ , taking expectations gives  $|J_{jk} - J_{jk}^*| \leq (2M_T^2 + 2M_T \tilde{C}_m) \varepsilon_\gamma$ , so  $\|\mathbf{J} - \mathbf{J}^*\|_{\text{op}} \leq C_J \varepsilon_\gamma$ . Combining bounds yields:

$$\|\mathbf{b}_\gamma\|_2 \leq \|\mathbf{S} - \mathbf{S}^*\|_2 + \|\mathbf{J} - \mathbf{J}^*\|_{\text{op}} \|\boldsymbol{\theta}^*\|_2 \leq (C_S + C_J M_\theta) \varepsilon_\gamma \equiv C \cdot \varepsilon_\gamma. \quad (20)$$

**Step 5: Population bound and finite-sample rate.** Since  $\mathbf{S} = \mathbf{J}\boldsymbol{\theta}^* + \mathbf{b}_\gamma$  and  $\boldsymbol{\theta}_\gamma \equiv \mathbf{J}^{-1}\mathbf{S}$ , we have  $\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^* = \mathbf{J}^{-1}\mathbf{b}_\gamma$ , so  $\|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 \leq \frac{\|\mathbf{b}_\gamma\|_2}{\lambda_{\min}(\mathbf{J})} \leq \frac{C\varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})}$ , establishing part (i). For part (ii): since  $\mathbf{J}\boldsymbol{\theta}_\gamma = \mathbf{S}$ ,  $\hat{\boldsymbol{\theta}}_\gamma - \boldsymbol{\theta}_\gamma = \hat{\mathbf{J}}^{-1}[(\hat{\mathbf{S}} - \mathbf{S}) - (\hat{\mathbf{J}} - \mathbf{J})\boldsymbol{\theta}_\gamma]$ . By Weyl's inequality,  $\lambda_{\min}(\hat{\mathbf{J}}) \geq \lambda_{\min}(\mathbf{J}) - \|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}} \geq \lambda_{\min}(\mathbf{J})/2$  on event  $\mathcal{E}_N$ , so  $\|\hat{\mathbf{J}}^{-1}\|_{\text{op}} \leq 2/\lambda_{\min}(\mathbf{J})$ . Lemma 2 gives  $\|\hat{\mathbf{S}} - \mathbf{S}\|_2 = \mathcal{O}_P(N^{-1/2} + r_N^2)$ , while non-orthogonality of the coupled Gram matrix yields  $\|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}} = \mathcal{O}_P(N^{-1/2} + r_N)$ . Combining with  $\|\boldsymbol{\theta}_\gamma\|_2 \leq M_\theta + C\varepsilon_\gamma/\lambda_{\min}(\mathbf{J})$  establishes the finite-sample bound in (ii). ■

**Proposition 2** (Exact Proxy Bias in the Symmetric Flip Model). *Let  $K = 2$ ,  $\boldsymbol{\pi}^* = (\frac{1}{2}, \frac{1}{2})$ ,  $T_t = V_t$  with  $V_t \perp (S_t, X_t, U_t)$ ,  $\mathbb{E}V_t = 0$ ,  $\text{Var}V_t = \sigma^2$ ,  $Y_t = \theta_{S_t}T_t + g_{S_t} + U_t$ ,  $\mathbb{E}[U_t \mid T_t, S_t] = 0$ , and  $\boldsymbol{\gamma}_t = (1 - \frac{\varepsilon}{2})\mathbf{H}_t + \frac{\varepsilon}{2}(\mathbf{1} - \mathbf{H}_t)$  with  $\varepsilon \in [0, 1)$  ( $\varepsilon_\gamma = \varepsilon$ ). Then  $\lambda_{\min}(\mathbf{J}) = \sigma^2(1 - \varepsilon)^2/2$  and*

$$\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^* = \frac{\varepsilon}{1 - \varepsilon} \cdot \frac{\theta_1 - \theta_2}{2} (1, -1)^\top, \quad \|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 = \frac{\varepsilon}{1 - \varepsilon} \frac{|\theta_1 - \theta_2|}{\sqrt{2}}. \quad (21)$$

*Proof.*  $m_k^\gamma = 0$  since  $\mathbb{E}V = 0$ ,  $V \perp S$ , so  $\tilde{T}_k = V$  and constants in  $\tilde{Y}_k$  drop out of  $\mathbf{S}$ . With  $a = 1 - \varepsilon/2$ ,  $b = \varepsilon/2$ ,  $\mathbf{M} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$ :  $\mathbf{J} = \frac{\sigma^2}{2}\mathbf{M}^2$  and  $\mathbf{S} = \frac{\sigma^2}{2}\mathbf{M}\boldsymbol{\theta}^*$ . Since  $\mathbf{M}$  has eigenvalues 1 (eigenvector  $(1, 1)^\top$ ) and  $1 - \varepsilon$  (eigenvector  $(1, -1)^\top$ ),  $\boldsymbol{\theta}_\gamma = \mathbf{J}^{-1}\mathbf{S} = \mathbf{M}^{-1}\boldsymbol{\theta}^*$ ,  $\lambda_{\min}(\mathbf{J}) = \sigma^2(1 - \varepsilon)^2/2$ , and  $(\mathbf{M}^{-1} - \mathbf{I})\boldsymbol{\theta}^*$  evaluates to the displayed vector. □

*Remark on Proxy Construction and Scope of Lower Bound.* In Proposition 2, the proxy  $\boldsymbol{\gamma}$  is constructed directly as a corrupted function of the latent state  $\mathbf{H}$  rather than inferred from observable features  $Z$ . An estimator having access to  $\boldsymbol{\gamma}$  directly could in principle invert the corruption matrix to recover  $\mathbf{H}$  consistently. Thus, Proposition 2 shows that the plug-in bound  $\mathcal{O}(\varepsilon_\gamma/\lambda_{\min}(\mathbf{J}))$  is attained by a corrupted proxy, but does not assert that calibrated posteriors inferred from  $Z$  necessarily incur this bias, nor do we make any minimax lower bound claim. The factorial experiment's synthetic proxy manipulation ( $\rho = 0.995$ ) shares this design.

## E POSTERIOR-ADJUSTED NUISANCES

Throughout,  $F_t = (X_t, \boldsymbol{\gamma}_t, \boldsymbol{\gamma}_t \otimes X_t)$ ,  $m^\gamma(F) = \mathbb{E}[T \mid F]$ ,  $\mu^\gamma(F) = \mathbb{E}[Y \mid F]$ ,  $\tilde{T} = T - m^\gamma$ ,  $\tilde{Y} = Y - \mu^\gamma$ , and the population system is  $J_{jk} = \mathbb{E}[\gamma_j \gamma_k \tilde{T}^2]$ ,  $S_k = \mathbb{E}[\gamma_k \tilde{T} \tilde{Y}]$ .

**Lemma 1 in detail.** For a direction  $\delta_m(F)$ ,  $\partial_t \mathbb{E}\psi_k(\boldsymbol{\theta}; m^\gamma + t\delta_m, \mu^\gamma)|_{t=0} = -\mathbb{E}[\gamma_k \delta_m \{\tilde{Y} - \sum_j \gamma_j \theta_j \tilde{T}\}] + \mathbb{E}[\gamma_k \tilde{T} \sum_j \gamma_j \theta_j \delta_m]$ . Conditioning on  $F$ , the first term is  $-\mathbb{E}[\gamma_k \delta_m (\mathbb{E}[\tilde{Y} \mid F] - \sum_j \gamma_j \theta_j \mathbb{E}[\tilde{T} \mid F])]$  = 0, and the second is  $\mathbb{E}[\gamma_k \sum_j \gamma_j \theta_j \delta_m \mathbb{E}[\tilde{T} \mid F]]$  = 0. The  $\mu$ -direction gives  $-\mathbb{E}[\gamma_k \delta_\mu \mathbb{E}[\tilde{T} \mid F]]$  = 0, and  $\partial_t \mathbb{E}[\gamma_j \gamma_k (\tilde{T} - t\delta_m)^2] = -2\mathbb{E}[\gamma_j \gamma_k \delta_m \mathbb{E}[\tilde{T} \mid F]]$  = 0. Nothing here uses calibration or the form of  $\boldsymbol{\gamma}$ : the weights only need to be functions of the conditioning set. For posterior-weighted regime nuisances  $m_k(X) = \mathbb{E}[\gamma_k T \mid X]/\mathbb{E}[\gamma_k \mid X]$ , the same

calculation leaves  $\mathbb{E}[\gamma_k \gamma_j \delta_m(X) \tilde{T}_k] \neq 0$  for soft weights, the  $\sqrt{\varepsilon_\gamma}$  sensitivity of Appendix D. *Numerical check* (`src/synthesis/coupled_law_check.py`,  $N = 10^6$ ): perturbing the treatment nuisance by  $t(1 + X^2 + \gamma)$ , the finite-difference derivative of the mean score is 0.4% of the Jacobian norm for posterior-adjusted nuisances (sampling noise) and 85% for posterior-weighted nuisances.

**Theorem 3 for posterior-adjusted nuisances.** Assume (i) calibration given  $F$ :  $P(S_t=k | F_t) = \gamma_{tk}$ , and (ii)  $(T, Y) \perp Z | (X, S)$ , so that  $\mathbb{E}[T | X, S, \gamma] = m_S(X)$ . Then  $m^\gamma(F) = \sum_k \gamma_k m_k(X)$  and  $|m^\gamma - m_S| \leq B_T \|\gamma - \mathbf{H}\|_1$ , and likewise for  $\mu^\gamma$ . Steps 3–5 of Appendix D then go through with the shared residual  $\tilde{T}$  in place of  $\tilde{T}_k$  and  $(\tilde{C}_m, \tilde{C}_\mu) = (B_T, B_Y)$ , giving  $\|\mathbf{b}_\gamma\|_2 \leq C\varepsilon_\gamma$ . For the stochastic part, Lemma 1 makes both  $\hat{\mathbf{S}}$  and  $\hat{\mathbf{J}}$  first-order insensitive to nuisance error, so  $\|\hat{\mathbf{S}} - \mathbf{S}\| + \|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}} = \mathcal{O}_P(N^{-1/2} + r_N^2)$  under purged cross-fitting, and the argument of Step 5 gives the stated rate with  $\rho_N = r_N^2$ . When (i) fails, the bound acquires the calibration error  $\mathbb{E}\|P(S \in \cdot | F) - \gamma\|_1$ .  $\square$

**Proof of Theorem 4.** Write  $r_t = S_t - \gamma_t$ ; by calibration  $\mathbb{E}[r | F] = 0$  and  $\mathbb{E}[r^2 | F] = v$ . Then  $\mathbb{E}[T | F] = m(X) + a\gamma$  and  $\tilde{T} = ar + V$ . Since  $\theta_S = \theta_0 + \Delta\theta S$  with  $\Delta\theta = \theta_1 - \theta_0$  and  $S^2 = S$ ,  $\mathbb{E}[\theta_S T | F] = \theta_0(m + a\gamma) + \Delta\theta\gamma(m + a)$ , so

$$\tilde{Y} = \theta_0 \tilde{T} + \Delta\theta\{(S - \gamma)(m + a) + SV\} + br + U.$$

For OR-DML,  $\mathbf{w}^\top \boldsymbol{\theta}^* = \theta_0 + \gamma\Delta\theta$ , and the residual at the truth is  $e = \tilde{Y} - (\theta_0 + \gamma\Delta\theta)\tilde{T} = r\{\Delta\theta(m + a(1 - \gamma)) + b\} + \Delta\theta rV + U = cr + \Delta\theta rV + U$ . Expanding  $\tilde{T}e = (ar + V)(cr + \Delta\theta rV + U)$  and conditioning on  $F$  (with  $V$  independent of  $(S, F)$ ,  $\mathbb{E}[U | S, F, V] = 0$ ) leaves only  $acv$ . Hence  $\mathbf{b}_\gamma = \mathbb{E}[\mathbf{w}\tilde{T}e] = a\mathbb{E}[\mathbf{w}cv]$  and, from  $\mathbb{E}[\tilde{T}^2 | F] = a^2v + \sigma^2$ ,  $\mathbf{J} = \mathbb{E}[\mathbf{w}\mathbf{w}^\top (a^2v + \sigma^2)]$ . For the pooled estimator ( $w \equiv 1$ ),  $\mathbb{E}[\tilde{T}\tilde{Y}] = \mathbb{E}[(\theta_0 + \gamma\Delta\theta)(a^2v + \sigma^2)] + a\mathbb{E}[cv]$ , and dividing by  $\mathbb{E}[\tilde{T}^2]$  gives the stated limit. Consistency of the sample version follows as in Appendix G, Steps 1–4, with remainder  $\mathcal{O}_P(N^{-1/2} + r_N^2 + \varepsilon_\tau)$  by Lemma 1. (a) is immediate. (b) is Appendix G. (c):  $\mathbf{J} - \mathbb{E}[\mathbf{w}\mathbf{w}^\top]\sigma^2 = a^2\mathbb{E}[\mathbf{w}\mathbf{w}^\top v] \succeq 0$  grows in  $|a|$  and in  $v$ .  $\square$

**Numerical check.** Table 4 compares the OR-DML estimate with the law on  $N = 10^6$  draws with a calibrated Beta posterior,  $m(X)$  linear and nuisances fitted on  $(1, X, \gamma, \gamma X)$ . The bias is hump-shaped in  $a$  (largest at  $a = 4$ ) while  $\lambda_{\min}(\mathbf{J})$  rises from 0.23 to 1.27, past the oracle value 0.50.

Table 4: Residual-state law for posterior-adjusted OR-DML: estimate versus law, true  $\boldsymbol{\theta}^* = (0.5, 0.5 + \Delta\theta)$  (`reports/synthesis/coupled_law_check.csv`).

| $a$ | $\Delta_g$ | $\Delta\theta$ | $\sigma$ | estimate       | law            | dev.   | $\lambda_{\min}(\mathbf{J})$ |
|-----|------------|----------------|----------|----------------|----------------|--------|------------------------------|
| 0   | 3          | 1              | 1        | (0.505, 1.500) | (0.500, 1.500) | 0.0048 | 0.227                        |
| 0.5 | 3          | 1              | 1        | (0.725, 1.708) | (0.723, 1.705) | 0.0029 | 0.231                        |
| 1   | 3          | 1              | 1        | (0.961, 1.882) | (0.953, 1.886) | 0.0085 | 0.244                        |
| 2   | 3          | 1              | 1        | (1.317, 2.094) | (1.317, 2.095) | 0.0008 | 0.292                        |
| 4   | 3          | 1              | 1        | (1.625, 2.091) | (1.624, 2.090) | 0.0010 | 0.487                        |
| 8   | 3          | 1              | 1        | (1.697, 1.874) | (1.695, 1.875) | 0.0013 | 1.266                        |
| 4   | -2         | 0.5            | 0.7      | (0.461, 0.648) | (0.462, 0.646) | 0.0018 | 0.370                        |
| 1.5 | 3          | 0              | 1        | (0.967, 0.967) | (0.969, 0.970) | 0.0026 | 0.264                        |
| 1   | 3          | 1              | 1        | (1.141, 2.019) | (1.146, 2.021) | 0.0048 | 0.114                        |

## F THE COUPLED SCORE UNDER STATE CONFUSION

### F.1 Proof of Proposition 1

With oracle regime-wise nuisances  $m_k(X) = \mathbb{E}[T | X, S=k]$  and  $\mu_k(X) = \mathbb{E}[Y | X, S=k]$ , the regime- $k$  residuals evaluated on an observation from regime  $s$  are  $\tilde{T}_k = (m_s - m_k) + \sigma_s V$  and  $\tilde{Y}_k = (\mu_s - \mu_k) + \theta_s \sigma_s V + U$ , where  $V$  has mean zero and unit variance and is independent of  $U$  and of  $S$ . The covariate parts cancel because the nuisances are regime-correct; only the regime-mean contrasts remain. Because  $\gamma_j$  is a function of  $s$  alone,

$$J_{jk} = \mathbb{E}[\gamma_j \gamma_k \tilde{T}_j \tilde{T}_k] = \sum_s \pi_s \gamma_j(s) \gamma_k(s) \mathbb{E}[\tilde{T}_j \tilde{T}_k | S=s] = \sum_s \pi_s \gamma_j(s) \gamma_k(s) [(m_s - m_j)(m_s - m_k) + \sigma_s^2],$$

and  $S_j = \mathbb{E}[\gamma_j \tilde{T}_j \tilde{Y}_j] = \sum_s \pi_s \gamma_j(s) [(m_s - m_j)(\mu_s - \mu_j) + \theta_s \sigma_s^2]$ . In the symmetric case put  $a = 1 - \alpha$ ,  $b = \alpha$ . The diagonal entries are  $\frac{1}{2}[a^2\sigma^2 + b^2(\Delta^2 + \sigma^2)]$ ; the off-diagonal entry is  $\frac{1}{2} \cdot 2ab\sigma^2 = ab\sigma^2$ , because on either regime the two residuals are  $\sigma V$  and  $\pm\Delta + \sigma V$ , whose product has mean  $\sigma^2$ . A symmetric  $2 \times 2$  matrix with equal diagonal

has eigenvectors  $(1, \pm 1)$  and eigenvalues “diagonal  $\pm$  off-diagonal”, which gives  $\lambda_{\pm}$  after using  $a + b = 1$ . For the displacement, write  $\mathbf{S} - \mathbf{J}\boldsymbol{\theta}^* = \mathbf{b}_{\gamma}$ , project  $\mathbf{b}_{\gamma}$  on  $(1, \pm 1)$  and divide by  $2\lambda_{\pm}$ . The  $(1, 1)$  component of  $\mathbf{b}_{\gamma}$  collects the leaked outcome contrast  $\alpha\Delta G$  and the quadratic leakage  $-\frac{1}{2}\alpha^2\Delta^2(\theta_0 + \theta_1)$ . The  $(1, -1)$  component collects the effect heterogeneity. The identity is verified symbolically in `src/synthesis/closed_form_factorial.py` (`displacement_formula_ok`). Setting  $\Delta = 0$  recovers the symmetric flip model of Proposition 2.  $\square$

**Proof of Corollary 1.** (a)  $\lambda_{-}(\varepsilon_{\gamma}) - \sigma^2/2 = \frac{1}{2}\sigma^2(\varepsilon_{\gamma}^2 - 2\varepsilon_{\gamma}) + \varepsilon_{\gamma}^2\Delta^2/8$ , which is positive iff  $\varepsilon_{\gamma}(\sigma^2 + \Delta^2/4) > 2\sigma^2$ . (b) Dropping the  $\mathcal{O}(\alpha^2)$  numerator term and the heterogeneity component,  $\|\boldsymbol{\theta}_{\gamma} - \boldsymbol{\theta}^*\|_2 \approx \sqrt{2}\alpha|\Delta G|/(\sigma^2 + \alpha^2\Delta^2)$  with  $\alpha = \varepsilon_{\gamma}/2$ . Maximizing  $\varepsilon_{\gamma}/(\sigma^2 + \varepsilon_{\gamma}^2\Delta^2/4)$  gives  $\varepsilon^* = 2\sigma/|\Delta|$  (also verified symbolically) and the stated peak. (c) The leading displacement is proportional to  $(1, 1)$ , the eigenvector of  $\lambda_{+}$ , so  $\|\mathbf{J}^{-1}\mathbf{b}_{\gamma}\| = \|\mathbf{b}_{\gamma}\|/\lambda_{+} \leq \|\mathbf{b}_{\gamma}\|/\lambda_{-}$ .  $\square$

## F.2 Verification against the factorial experiment

Table 5 reports, for every cell of the  $7 \times 7$  factorial with  $\varepsilon_{\gamma} > 0$ , the ratio of the observed median error (100 replications) to the population error from Proposition 1, using the factorial’s own parameters: stationary weights  $(0.556, 0.444)$ ,  $m = (2, 5)$ ,  $g = (15, 50)$ ,  $\theta = (0.75, 2.5)$ ,  $\sigma_0 = 1$ . All 42 ratios lie in  $[0.98, 1.02]$ , and the population  $\lambda_{\min}$  matches the mean realized value to within 1.1%. The finite-sample component is visible only at  $\varepsilon_{\gamma} = 0$ , where the population error is zero.

Table 5: Observed median error divided by the exact population error (Proposition 1), factorial design.

| $\sigma_1 \setminus \varepsilon_{\gamma}$ | 0.01  | 0.02  | 0.05  | 0.1   | 0.2   | 0.4   |
|-------------------------------------------|-------|-------|-------|-------|-------|-------|
| 0.08                                      | 0.984 | 0.986 | 0.998 | 0.998 | 0.999 | 1.001 |
| 0.15                                      | 0.987 | 0.985 | 0.989 | 1.000 | 0.999 | 1.000 |
| 0.3                                       | 0.992 | 0.989 | 0.987 | 0.990 | 1.000 | 1.000 |
| 0.5                                       | 1.000 | 0.994 | 0.991 | 0.990 | 0.995 | 0.999 |
| 0.75                                      | 1.006 | 0.997 | 0.995 | 0.994 | 0.996 | 1.000 |
| 1                                         | 1.008 | 1.014 | 1.017 | 1.011 | 1.007 | 1.009 |
| 1.5                                       | 1.014 | 1.012 | 1.018 | 1.021 | 1.020 | 1.022 |

## F.3 Phantom resolution beyond the coupled score

**Hard classifiers.** Let  $J_{\text{true}} = \mathbb{E}[\tilde{T}_t^2 H_t H_t^{\top}]$  with true residual treatment variances  $a > b > 0$  in a balanced two-state model. A symmetric hard classifier that flips labels with probability  $p \in (0, 1/2)$  yields inferred diagonal information proportional to  $((1-p)a + pb, pa + (1-p)b)/2$ . The smaller inferred value  $[pa + (1-p)b]/2$  exceeds  $b/2$ , and at  $p = 1/2$  the apparent condition number is one however large the oracle condition number  $a/b$  is. In a three-state Monte Carlo with residual treatment standard deviations  $(1, 0.3, 0.08)$  and  $\Delta_Z = 0.5$ , oracle  $\lambda_{\min} \approx 0.0022$ , while the soft and hard generated-state values are about 0.0246 and 0.0457 ( $11.4\times$  and  $21.2\times$  inflation; research ledger in the supplementary archive).

**Why posterior means are insufficient for information moments.** For one-hot  $H_t$  and posterior  $q_t = P(S_t | \mathcal{I}_t)$ ,  $\mathbb{E}[H_t H_t^{\top} | \mathcal{I}_t] = \text{diag}(q_t)$ , not  $q_t q_t^{\top}$ . For any weight  $W_t$  measurable with respect to  $\mathcal{I}_t$ ,  $\mathbb{E}[W_t H_t H_t^{\top}] = \mathbb{E}[W_t \text{diag}(q_t)]$ . This can recover oracle second moments under a correctly specified posterior whose information set contains every state-informative variable entering  $W_t$ . It is not a free correction, though: misspecifying the state-treatment model recreates phantom resolution, so we treat completed geometry as a sensitivity diagnostic.

**Directional reliability.** The bound  $\|\mathbf{b}_{\gamma}\|/\lambda_{\min}(\mathbf{J})$  is worst-directional. For a scientific contrast  $\mathbf{c}$  the relevant first-order quantity is  $|\mathbf{c}^{\top} \mathbf{J}^{-1} \mathbf{b}_{\gamma}|$ . In  $K \geq 3$  regimes, equal-magnitude perturbations can therefore have very different consequences depending on their alignment with the information eigenvectors, and no task-agnostic scalar representation score is sufficient in general. Fine latent states may still be needed for nuisance adjustment even when a separate coefficient for every state is unsupported, which is why we distinguish adjustment resolution from target resolution.

## G THE RESIDUAL-STATE BIAS LAW

**Setting.**  $S_t \in \{0, \dots, K-1\}$  with indicator vector  $D_t \in \{0, 1\}^{K-1}$ ;  $T_t = m(X_t) + a^\top D_t + V_t$  and  $Y_t = \theta T_t + g(X_t) + b^\top D_t + U_t$ ;  $\hat{\gamma}_t = \hat{\gamma}_t(Z_{1:N})$  is any posterior computed from the proxies; nuisances are cross-fitted on  $F_t = (X_t, \hat{\gamma}_t)$  with purged blocks;  $\hat{\theta} = \sum \tilde{T}_t \tilde{Y}_t / \sum \tilde{T}_t^2$ .

**Assumption 5** (Exogenous noise).  $V_t$  is independent of  $(S, Z, X)$  at all times,  $\mathbb{E}V_t = 0$ ,  $\text{Var}V_t = \sigma^2 > 0$ ;  $\mathbb{E}[U_t | S, Z, X, V] = 0$ ; both noises have summable autocovariances and bounded  $4+\delta$  moments.

**Assumption 6** (Posterior as a local feature). The process is stationary and  $\alpha$ -mixing;  $\hat{\gamma}_t$  depends on  $Z_s$  for  $|s - t| > \tau$  only through terms of size  $\varepsilon_\tau$  in  $L^2$  (filter stability).

**Assumption 7** (Nuisance learners). The cross-fitted learners for  $\mathbb{E}[T | F]$  and  $\mathbb{E}[Y | F]$  have  $L^2$  errors  $r_T, r_Y$  ( $r_h = r_Y + |\theta|r_T$ ), and the embargo exceeds the mixing time.

**Pooled law for  $K \geq 2$  regimes.** For homogeneous effects and  $w \equiv 1$ , Theorem 4 extends to  $K \geq 2$ : posterior-feature DML satisfies  $\hat{\theta} - \theta = a^\top \Sigma_N b / (a^\top \Sigma_N a + \sigma^2) + O_P(N^{-1/2} + r_T r_h + \varepsilon_\tau)$ , where  $D_t$  is the  $(K-1)$ -vector of regime indicators,  $\tilde{D}_t = D_t - \mathbb{E}[D_t | F_t]$ ,  $\Sigma_N = N^{-1} \sum_t \tilde{D}_t \tilde{D}_t^\top$ , and  $T = m(X) + a^\top D + V$ ,  $Y = \theta T + g(X) + b^\top D + U$ . Calibration is not needed for this form, because  $\Sigma_N$  is defined relative to the conditional mean given  $F$  (Remark: the relevant  $\Sigma_N$  is that of the learner's projection when the learner class is restricted). *Proof. Step 0 (exact decomposition).* By Assumption 5,  $\mathbb{E}[T_t | F_t] = m(X_t) + a^\top c_t$  and  $\mathbb{E}[Y_t | F_t] = \theta \mathbb{E}[T_t | F_t] + g(X_t) + b^\top c_t$  with  $c_t = \mathbb{E}[D_t | F_t]$ . With nuisance errors  $\delta_t^T, \delta_t^Y$  and  $\delta^h = \delta^Y - \theta \delta^T$ ,  $\tilde{T}_t = a^\top \tilde{D}_t + V_t - \delta_t^T$  and  $\tilde{Y}_t - \theta \tilde{T}_t = b^\top \tilde{D}_t + U_t - \delta_t^h$ , and  $\hat{\theta} - \theta = \sum_t \tilde{T}_t (\tilde{Y}_t - \theta \tilde{T}_t) / \sum_t \tilde{T}_t^2$  exactly. *Step 1.*  $N^{-1} \sum (a^\top \tilde{D}_t)(b^\top \tilde{D}_t) = a^\top \Sigma_N b$ ,  $N^{-1} \sum (a^\top \tilde{D}_t)^2 = a^\top \Sigma_N a$ , and  $N^{-1} \sum V_t^2 = \sigma^2 + O_P(N^{-1/2})$ . *Step 2.* Conditionally on  $\mathcal{G} = \sigma(S, Z, X)$ ,  $\sum_t V_t b^\top \tilde{D}_t$ ,  $\sum_t a^\top \tilde{D}_t U_t$  and  $\sum_t V_t U_t$  have mean zero and variance  $O(N)$ , hence are  $O_P(N^{1/2})$ . *Step 3.* Within a fold,  $\delta_t = \delta(F_t)$  is fixed given the training folds and  $\mathbb{E}[\tilde{D}_t | F_t] = 0$ . So  $\sum \delta_t \tilde{D}_t$ ,  $\sum \delta_t V_t$  and  $\sum \delta_t U_t$  have conditional mean  $O(\varepsilon_\tau)$  and variance  $\leq CN \|\delta\|_{L^2}^2$  by Davydov's inequality; and  $|\sum \delta_t^T \delta_t^h| \leq N r_T r_h$ . *Step 4.* Write the ratio as  $(A + e_1)/(B + e_2)$  with  $A = a^\top \Sigma_N b$ ,  $B = a^\top \Sigma_N a + \sigma^2 \geq \sigma^2$  and  $e_i = O_P(N^{-1/2} + r_T r_h + \varepsilon_\tau)$ . Then  $(A + e_1)/(B + e_2) - A/B = (e_1 - (A/B)e_2)/(B + e_2)$ , with  $|A/B| \leq \sqrt{b^\top \Sigma_N b} / (2\sigma)$  bounded.  $\square$

**Hump, sup bound and monotonicity ( $K = 2$ ).** (i)  $a^2 v + \sigma^2 \geq 2|a|\sqrt{v}\sigma$ , with equality at  $|a| = \sigma/\sqrt{v}$ , so  $|B| \leq |\Delta_g| \sqrt{v} / (2\sigma)$ . (ii) With  $\lambda = \text{Var}(T) = a^2 v + \sigma^2$ ,  $|B| = |\Delta_g| \sqrt{v} \sqrt{\lambda - \sigma^2} / \lambda$ , which increases in  $\lambda$  on  $[\sigma^2, 2\sigma^2]$  and decreases afterwards, while  $\lambda$  increases monotonically in  $|a|$ . Conditioning therefore cannot be a monotone certificate. (iii)  $\partial_v[v/(a^2 v + \sigma^2)] = \sigma^2/(a^2 v + \sigma^2)^2 > 0$ . For  $\mathcal{F} \subseteq \mathcal{F}'$ , the law of total variance gives  $\mathbb{E}\text{Cov}(D | \mathcal{F}') \preceq \mathbb{E}\text{Cov}(D | \mathcal{F})$ ; a hard label is a measurable function of  $\hat{\gamma}$ , so the soft feature is never worse for  $K = 2$ .  $\square$

**Observable plug-in.** Let  $CE = \mathbb{E}\|\hat{\gamma}_t - \mathbb{E}[D_t | F_t]\|^2$ . With  $c = \mathbb{E}[D | F]$ ,  $\gamma(1 - \gamma) - c(1 - c) = (\gamma - c)(1 - 2c) - (\gamma - c)^2$ , so  $|\mathbb{E}\hat{v} - v| \leq \sqrt{CE} + CE$ . If  $m$  is linear and the posterior calibrated, the OLS coefficient  $\hat{a}$  of  $\hat{\gamma}$  in the regression of  $T$  on  $(1, X, \hat{\gamma})$  is consistent for  $a$ , and  $\sigma^2 = \text{Var}(\tilde{T}) - a^\top \Sigma a$ .  $\square$

**Heterogeneous effects ( $K = 2$ ).** With  $\theta(S) \in \{\theta_0, \theta_1\}$ ,  $\bar{\theta} = \mathbb{E}\theta(S)$ ,  $T_t = \bar{m} + m_0(X_t) + aD_t + V_t$ ,  $\mathbb{E}m_0 = 0$  and  $\mathbb{E}[m_0(X)\text{Var}(D | F)] = 0$ ,  $\hat{\theta} \rightarrow_p \bar{\theta} + av[b + a(\theta_1 - \bar{\theta}) + \bar{m}(\theta_1 - \theta_0)]/(a^2 v + \sigma^2)$ . Expanding  $\mathbb{E}[\tilde{T}\theta(S)T]$  with  $\tilde{T} = a\tilde{D} + V$  gives  $a\bar{m}(\theta_1 - \theta_0)v + a^2\theta_1 v + \sigma^2\bar{\theta}$ ; subtract  $\bar{\theta}(a^2 v + \sigma^2)$ . This is the form used for Table 2's semi-synthetic prediction and for the frontier DGP.  $\square$

**Learned HMM parameters.** *Setting (E).* The chain has finitely many states with  $A_{ij}^* \geq \varepsilon^* > 0$ ; the emissions are Gaussian with parameters in a compact set and variances bounded below; and  $\vartheta^*$  is interior, identifiable up to labels, with nonsingular Fisher information. Let  $\gamma_t^*$  be the smoothing posterior at  $\vartheta^*$  and  $\hat{\gamma}_t$  at  $\hat{\vartheta}$ . (H1)  $\|\hat{\vartheta} - \vartheta^*\| = O_P(N^{-1/2})$  (and  $\mathbb{E}\|\sqrt{N}(\hat{\vartheta} - \vartheta^*)\|^4 \leq C$  for statements in expectation). This is a hypothesis on the output of EM, multistart checks optimizer stability, not this asymptotic rate.

**Lemma 3** (Derivative of the smoother). *Under (E) there are  $r > 0$ ,  $C < \infty$  with  $D_t := \sup_{\|\vartheta - \vartheta^*\| \leq r} \|\partial_{\vartheta} \gamma_{t, \vartheta}\| \leq C \sum_s \kappa^{|t-s|} (1 + \|Z_s\|^2)$ ,  $\kappa = 1 - \varepsilon^*/2$ , and  $|\gamma_t^* - \gamma_t^{*,(m)}| \leq \kappa^m$  for the window- $m$  posterior; hence  $\sup_{t,N} \mathbb{E}D_t^p < \infty$  and  $N^{-1} \sum_t D_t^2 = O_P(1)$ .*

*Proof.* Given  $Z$ , the hidden chain has kernels  $K_s(i, j) \geq (\varepsilon^*/2)w_j / \sum_l w_l$ , a Doeblin minorisation independent of the emissions, so its Dobrushin coefficient is at most  $\kappa$  and conditional covariances decay as  $\kappa^{|t-s|}$ . Fisher's identity writes  $\partial_{\vartheta} \gamma_t(k) = \sum_s \text{Cov}(\mathbb{I}\{S_t = k\}, h_s | Z)$  with  $\text{osc}(h_s) \leq C(1 + \|Z_s\|^2)$ ; summing gives the bound, and the same contraction gives forgetting.  $\square$

**Proposition 3** (Rate for learned parameters). *Assume (E) and (H1), and let  $\delta_X = N^{-1} \sum_t \mathbb{E}[\mathbb{E}[D_t | Z, X] - \gamma_t^*]^2$  be the regime information carried by the controls beyond the proxies. Then  $N^{-1} \sum_t \|\hat{\gamma}_t - \gamma_t^*\|^2 = O_P(N^{-1})$ ,  $|\mathbb{E}\hat{v} - v^*| = O(N^{-1/2})$ ,  $CE \leq (\sqrt{\delta_X} + O(N^{-1/2}))^2$  and  $|v_N - v^*| \leq 2\sqrt{\delta_X} + O(N^{-1/2})$ . The plug-in is therefore  $N^{-1/2}$ -accurate when  $\delta_X = 0$ , and its guarantee degrades by  $\sqrt{\delta_X}$  otherwise.*

*Proof.* On  $\{\|\hat{\vartheta} - \vartheta^*\| \leq r\}$  the mean-value theorem and Lemma 3 give  $\|\hat{\gamma}_t - \gamma_t^*\| \leq D_t \|\hat{\vartheta} - \vartheta^*\|$ , and averaging gives the first claim. Since  $|x(1-x) - y(1-y)| \leq |x-y|$  the second follows. With  $P$  the  $L^2$  projection onto  $\sigma(X, \hat{\gamma})$ ,  $CE = \|P(\hat{\gamma} - \gamma^X)\|^2 \leq \|\hat{\gamma} - \gamma^X\|^2$ , and  $|v_N - v^*| \leq \|c - \gamma^*\| \leq \|\hat{\gamma} - \gamma^X\| + \sqrt{\delta_X}$ .  $\square$  *Empirical checks* (supplement, `reports/bias_law/x21-x30`). When the emission model is correct,  $\mathbb{E}|\hat{\gamma} - \gamma^*|$  decays with log-log slope  $-0.49$ . Under misspecified (correlated  $t_5$ ) emissions the plug-in  $\hat{v}$  equals the pseudo-true value but understates the realized residual variance by a factor of two, in that experiment. This is not a general lower bound: a true constant posterior 0.9 replaced by 0.5 gives plug-in Gini 0.25 versus true residual variance 0.09, whereas reversing the posteriors understates it. Refitting the HMM on training folds only leaves the bias unchanged (correlation 0.999 between fold-wise and global estimates). Mumbai’s likelihood is multimodal: five restarts span 1,182 log-likelihood units, so (H1) cannot be assumed for arbitrary restart outputs, and the best restart changes its robustness value from 11.9 to 7.1 SD without changing any conclusion. On the real series  $\delta_X > 0$ , because the controls contain the proxy variables.

**A priori residual variance.** For a continuous AR(1) latent state with coefficient  $\phi$  observed with information rate  $\iota = h^\top R^{-1}h$ , the Kalman-smoothed residual variance is  $v = (1 - \phi^2)/\sqrt{(1 + \phi^2 + \iota(1 - \phi^2))^2 - 4\phi^2}$  (Wiener–Kolmogorov). The worst-case bias therefore decays only as  $\iota^{-1/4}$  while  $\iota \ll (1 - \phi^2)^{-1}$ : halving it needs about  $16\times$  the sensor information. For a persistent two-state chain with Mahalanobis separation  $d$  and switch rate  $r$ , residual variance is paid at change points,  $v \approx cre^{-d^2/8}/d$  with  $c \rightarrow \sqrt{\pi/2}$  for an isolated switch. Discrete regimes with informative proxies are therefore far less dangerous than a continuous latent state with the same sensor information. Statements, proofs, numerical checks and fitted constants are in the supplementary archive (`archive/manuscript_lineages/v2.bias_law_main.tex`, `reports/bias_law/x6`, `x15-x20`, `x28`, `x31`).

## G.1 Experiments for the bias law

All use `src/bias_law/`: a persistent chain (stay probability 0.9), 2-d Gaussian proxies,  $N = 3000$ ,  $\Delta_g = 3$ , an EM-fitted HMM, five purged folds with  $\tau = 24$ , and 12 replications per cell. Across 72 two-regime cells (smoother and filter; linear and boosted nuisances;  $\Delta_Z \in \{0.5, 0.8, 1.2\}$ ; eight values of  $\Delta_T$  from 0.25 to 32), the law with the observable plug-in  $\hat{v}$  has correlation 0.996 and RMSE 0.012 with the measured bias, and median relative error 5.2% on cells with  $|\text{bias}| > 0.05$  (3.5% with the realized  $v_N$ ; Table 6). Subtracting the plug-in bias with the true  $b$  reduces mean absolute error from 0.126 to 0.021. Table 7 shows the pitfall behind “conditioning sweeps”: if nuisances omit the regime,  $\lambda_{\min}$  grows from 0.54 to 14.4 while error is hump-shaped (peak 14.0); with regime-aware nuisances  $\lambda_{\min}$  is flat at 0.51 and the error stays in  $[0.06, 0.10]$ . Table 8 shows the law explaining the frontier DGP of Table 2. Table 9 shows that temperature scaling changes  $\hat{v}$  by a factor of 14 while the realized bias is constant.

Table 6: Agreement between measured bias and the law (bias-law experiments E1–E2).

| Setting             | cells | $\text{corr}(\hat{\theta} - \theta, \text{law}(\hat{v}))$ | RMSE  | med. rel. err. | $\text{corr}(\text{law}(v_N))$ |
|---------------------|-------|-----------------------------------------------------------|-------|----------------|--------------------------------|
| $K=2$ , filter, lin | 24    | 0.997                                                     | 0.016 | 7.3%           | 1.000                          |
| $K=2$ , smooth, gbm | 24    | 0.998                                                     | 0.011 | 4.3%           | 0.995                          |
| $K=2$ , smooth, lin | 24    | 0.998                                                     | 0.009 | 5.1%           | 0.999                          |
| $K=3$ , smooth, lin | 8     | 0.999                                                     | 0.010 | 9.8%           | 0.998                          |

## H INFERENCE

**Scope.** This proof is for the *posterior-adjusted coupled* estimator. Its target is the regularized proxy parameter, not automatically the structural effect. Full-sample HMM smoothing in the empirical analyses is outside this theorem. Fold-wise and global estimates having correlation 0.999 is a stability observation, not a justification of their standard errors. Orthogonality in  $m, \mu$  does not imply orthogonality in the fitted representation.

**Assumption 8** (Sufficient conditions for posterior-adjusted inference). (D) For a fixed representation parameter

Table 7: Conditioning sweep on the frontier DGP ( $\Delta_Z = 1.5$ ): the role of regime-aware nuisances.

| $\Delta_T$ | nuisances on $X$ only     |                               | nuisances on $(X, S)$     |                               |
|------------|---------------------------|-------------------------------|---------------------------|-------------------------------|
|            | $\lambda_{\min}(\hat{J})$ | $\ \hat{\theta} - \theta\ _2$ | $\lambda_{\min}(\hat{J})$ | $\ \hat{\theta} - \theta\ _2$ |
| 0.4        | 0.54                      | 5.92                          | 0.51                      | 0.10                          |
| 0.8        | 0.61                      | 10.34                         | 0.51                      | 0.08                          |
| 1.5        | 0.86                      | 14.05                         | 0.51                      | 0.07                          |
| 3.0        | 1.88                      | 13.38                         | 0.51                      | 0.07                          |
| 6.0        | 5.61                      | 8.76                          | 0.51                      | 0.07                          |
| 10.0       | 14.38                     | 5.73                          | 0.51                      | 0.06                          |

 Table 8: Frontier DGP (30 replications,  $N = 1200$ ): excess bias of soft conditioning over the oracle state, and the law.

| $\Delta_Z$ | DML on $X$ | soft $\hat{\gamma}$ | oracle $S$ | excess bias | law( $v_N$ ) | law( $\hat{v}$ ) |
|------------|------------|---------------------|------------|-------------|--------------|------------------|
| 0.2        | 10.02      | 9.773               | 1.535      | 8.238       | 8.236        | 7.418            |
| 0.5        | 10.02      | 7.776               | 1.535      | 6.241       | 6.267        | 5.655            |
| 1.0        | 10.02      | 3.796               | 1.535      | 2.260       | 2.299        | 1.891            |
| 1.5        | 10.02      | 1.923               | 1.535      | 0.388       | 0.402        | 0.320            |
| 2.0        | 10.02      | 1.559               | 1.535      | 0.024       | 0.024        | 0.016            |
| 4.0        | 10.02      | 1.535               | 1.535      | 0.000       | 0.000        | 0.000            |

$\Lambda$ , the augmented evaluation process  $(Y_t, T_t, F_t, \gamma_t)$  is strictly stationary and geometrically *absolutely regular*, with  $\beta(h) \leq M \exp(-c_\beta h)$ . This condition concerns the representation as well as the raw observations; it does not follow merely by assuming  $\alpha$ -mixing of the latter. Use a fixed number of contiguous evaluation folds of size proportional to  $N$ , and separate their nuisance-training observations, and the historical calibration window, by  $\tau_N \geq C \log N$ ,  $C > 1/c_\beta$ . Any finite-history approximation to a filter must have an additional aggregate score error  $o_P(N^{-1/2})$ , and its history must be included in the purge.

(R) Treatment, outcome, and fitted nuisances have uniform bounded envelopes (nuisances may be clipped). For the conditional targets  $m^\Lambda(F) = \mathbb{E}[T | F]$ ,  $\mu^\Lambda(F) = \mathbb{E}[Y | F]$ , held-out nuisance errors satisfy  $r_N = \max_{\text{folds}}(\|\hat{m} - m^\Lambda\|_2 + \|\hat{\mu} - \mu^\Lambda\|_2) = o_P(N^{-1/4})$ , uniformly in a neighborhood of  $\Lambda^*$ . The fixed- $K$  population matrix  $B_\lambda = J + \lambda I$  has smallest eigenvalue bounded away from zero. Here  $\lambda$  is fixed; a deterministic convergent sequence may be used if stability and the following expansion hold uniformly.

(H) For an estimated representation, a separated calibration sample of length  $N_{\text{cal}}$  gives an asymptotically linear regular MLE with  $\sqrt{N_{\text{cal}}}(\hat{\Lambda} - \Lambda^*) \Rightarrow \mathcal{N}(0, \mathcal{I}_\Lambda^{-1})$  and  $N/N_{\text{cal}} \rightarrow \kappa$ . The population moment map below is differentiable, and its empirical version is stochastically differentiable with remainder  $o_P(N^{-1/2})$  at  $\hat{\Lambda}$ . These are additional conditions on the filter and its population nuisance targets, not consequences of nuisance orthogonality or of running EM (Leroux, 1992; Bickel et al., 1998; Douc et al., 2004). For a known representation  $\kappa = 0$  and (H) is unnecessary.

For fixed  $\Lambda$ , put  $r = T - m^\Lambda(F)$ ,  $q = Y - \mu^\Lambda(F)$  and

$$\Psi_t(\theta; \Lambda) = \gamma_t r_t (q_t - r_t \gamma_t^\top \theta), \quad J = \mathbb{E}[\gamma \gamma^\top r^2], \quad S = \mathbb{E}[\gamma r q].$$

Let  $\theta_\lambda = B_\lambda^{-1} S$ . At this target  $\mathbb{E}\Psi_t(\theta_\lambda) = \lambda \theta_\lambda$ , so the CLT and HAC covariance use  $\tilde{\Psi}_t = \Psi_t - \mathbb{E}\Psi_t$ .

*Proof of Theorem 5. Step 1 (nuisance expansion).* Write  $\delta_m = \hat{m} - m^\Lambda$  and  $\delta_\mu = \hat{\mu} - \mu^\Lambda$ , and  $h = \gamma^\top \theta_\lambda$ . The exact score difference on a held-out fold is

$$\hat{\Psi} - \Psi = \gamma \{-\delta_m q - r \delta_\mu + 2hr \delta_m \delta_\mu - h \delta_m^2\}.$$

The first three terms have zero population mean because  $r, q$  are centered conditional on  $F$  and  $\gamma, h, \delta_m, \delta_\mu$  are  $F$ -measurable. The last two have mean bounded by  $C(\|\delta_m\|_2 \|\delta_\mu\|_2 + \|\delta_m\|_2^2) = O_P(r_N^2)$ . No posterior-weighted Gram condition (G) is needed.

*Step 2 (dependence and held-out errors).* Absolute-regularity coupling of separated blocks (Yu, 1994) replaces training and evaluation blocks by independent copies with total failure probability at most  $O(N\beta(\tau_N)) = o(1)$ .

Table 9: Temperature scaling of the posterior ( $K = 2$ , boosted nuisances, 20 replications,  $\Delta_Z = 0.8$ ,  $\Delta_T = 2$ ).

| $\tau$ | bias  | law( $\hat{v}$ ) | law( $v_N$ ) | $\hat{v}$ | $v_N$ | $\sqrt{CE} + CE$ |
|--------|-------|------------------|--------------|-----------|-------|------------------|
| 0.30   | 0.128 | 0.033            | 0.160        | 0.006     | 0.030 | 0.071            |
| 0.50   | 0.127 | 0.056            | 0.160        | 0.010     | 0.030 | 0.053            |
| 0.75   | 0.127 | 0.088            | 0.160        | 0.016     | 0.030 | 0.035            |
| 1.00   | 0.127 | 0.123            | 0.160        | 0.022     | 0.030 | 0.025            |
| 1.50   | 0.127 | 0.201            | 0.160        | 0.039     | 0.030 | 0.038            |
| 2.50   | 0.127 | 0.355            | 0.160        | 0.077     | 0.030 | 0.098            |

 Table 10: When can  $b$  be learned? Posterior-DML versus a correctly specified joint Markov-switching MLE and proximal 2SLS with a negative-control outcome (bias, RMSE in parentheses).

| Scenario                                | $\Delta_T$ | Posterior-DML | Joint MLE    | Proximal 2SLS |
|-----------------------------------------|------------|---------------|--------------|---------------|
| HMM correctly specified                 | 1          | +0.20 (0.20)  | +0.00 (0.02) | -0.01 (0.02)  |
| HMM correctly specified                 | 4          | +0.40 (0.40)  | +0.00 (0.02) | +0.02 (0.06)  |
| nonlinear $f(X)$ (GBM nuisances)        | 1          | +0.16 (0.17)  | -0.01 (0.01) | -0.00 (0.04)  |
| nonlinear $f(X)$ (GBM nuisances)        | 4          | +0.37 (0.37)  | -0.00 (0.01) | -0.03 (0.06)  |
| heavy-tailed ( $t_3$ ) proxy noise      | 1          | +0.15 (0.16)  | +0.01 (0.02) | +0.00 (0.04)  |
| heavy-tailed ( $t_3$ ) proxy noise      | 4          | +0.32 (0.32)  | -0.00 (0.01) | -0.04 (0.05)  |
| true $K=3$ , fitted $K=2$               | 1          | +0.46 (0.47)  | -0.00 (0.02) | -0.01 (0.05)  |
| true $K=3$ , fitted $K=2$               | 4          | +0.56 (0.57)  | +0.00 (0.01) | +0.06 (0.16)  |
| heterogeneous $\theta(S)$ (ATE target)  | 1          | +0.26 (0.26)  | -0.01 (0.03) | -0.02 (0.06)  |
| heterogeneous $\theta(S)$ (ATE target)  | 4          | +0.66 (0.66)  | -0.00 (0.02) | -0.01 (0.08)  |
| negative control leaks ( $W \ni 0.3T$ ) | 1          | +0.21 (0.21)  | +0.01 (0.03) | -0.45 (0.45)  |
| negative control leaks ( $W \ni 0.3T$ ) | 4          | +0.39 (0.39)  | -0.00 (0.01) | -0.46 (0.47)  |
| continuous AR(1) latent, SNR 0.5        | 1          | +0.83 (0.84)  | +0.79 (0.80) | -0.02 (0.04)  |
| continuous AR(1) latent, SNR 1          | 1          | +0.77 (0.78)  | +0.79 (0.79) | -0.02 (0.04)  |
| continuous AR(1) latent, SNR 0.5        | 2          | +0.90 (0.90)  | +0.89 (0.89) | -0.03 (0.08)  |
| continuous AR(1) latent, SNR 1          | 2          | +0.87 (0.88)  | +0.88 (0.89) | -0.02 (0.06)  |

An  $\alpha$ -mixing bound alone does not provide this total-variation coupling for arbitrary learned fits. Conditional on an independent training copy, each bounded linear perturbation in Step 1 has second moment  $O(r_N^2)$  and covariance at lag  $h$  bounded by  $\min\{Cr_N^2, C\beta(h)\}$ . Summing gives  $O(r_N^2(1 + |\log r_N|)) = o_P(1)$  for the variance of its normalized fold sum. The empirical quadratic terms are  $O_P(r_N^2)$  by Cauchy-Schwarz and conditional Markov bounds. Summing over the fixed number of folds yields

$$\hat{S} - \hat{J}\theta_\lambda = N^{-1} \sum_t \Psi_t(\theta_\lambda; \Lambda) + o_P(N^{-1/2}).$$

The same argument gives  $\hat{J} \rightarrow_p J$ . This is a sufficient-condition result for clipped, bounded nuisances, not a claim for arbitrary learners under mixing alone.

*Step 3 (exact normal equation and CLT).* The sample system is  $(\hat{J} + \lambda I)\hat{\theta}_\lambda = \hat{S}$ . Subtracting  $(J + \lambda I)\theta_\lambda = S$  gives

$$\sqrt{N}(\hat{\theta}_\lambda - \theta_\lambda) = B_\lambda^{-1} N^{-1/2} \sum_t \tilde{\Psi}_t + o_P(1).$$

The centered, stationary, bounded geometrically  $\beta$ -mixing score obeys the mixing CLT (Ibragimov, 1962), with long-run covariance  $\Omega_\lambda = \sum_{h \in \mathbb{Z}} \text{Cov}(\Psi_0, \Psi_h)$ .

*Step 4 (representation estimation).* At fixed  $\theta_\lambda$  define the  $K$ -by- $\dim(\Lambda)$  total derivative

$$A_\Lambda = \partial_{\Lambda^\top} \mathbb{E}[\Psi_t(\theta_\lambda; \Lambda)]|_{\Lambda=\Lambda^*}.$$

This is the derivative of the coupled vector moment, including both occurrences of the weights and the posterior-dependent conditional targets. Under (H), the expansion gains  $B_\lambda^{-1} A_\Lambda \sqrt{N}(\hat{\Lambda} - \Lambda^*)$ . Separation from the calibration sample gives asymptotic independence under (D), hence

$$\Sigma_\lambda^{\text{tot}} = B_\lambda^{-1} \{\Omega_\lambda + \kappa A_\Lambda \mathcal{I}_\Lambda^{-1} A_\Lambda^\top\} B_\lambda^{-T}.$$

For same-sample representation fitting, independence fails and the joint influence covariance must also include cross terms; that setting is not proved here.

*Step 5 (structural centering).* The exact displacement is  $\theta_\lambda - \theta^* = B_\lambda^{-1}(b_\gamma - \lambda\theta^*)$ . A structural Wald interval requires this displacement to be  $o(N^{-1/2})$ , with a nondegenerate limiting covariance. The bound in the main theorem is a sufficient condition; nuisance orthogonality alone cannot remove  $b_\gamma$ .  $\square$

**What the code estimates.** `src/or_dml.py` now constructs the observation-level moment matrix used in its solve, centers its resulting scores, and uses their HAC sandwich. `src/synthesis/estimators.py` implements the same coupled vector score. These routines *do not estimate*  $A_\Lambda \mathcal{I}_\Lambda^{-1} A_\Lambda^\top$  or the HMM joint influence covariance. Their intervals are conditional first-stage proxy-target diagnostics, not implementations of the full variance in Theorem 5. A residual block bootstrap that keeps the fitted HMM and nuisance residuals fixed has the same limitation. HMM-uncertainty inference requires the additional influence calculation above or a justified full-pipeline refit scheme; misspecification and structural bias require separate sensitivity analysis. Forward state updating with globally fitted parameters is likewise not a held-out calibration fit.

**Legacy weighted design.** Its nonzero first-order treatment-nuisance derivative requires an additional rate such as  $\sqrt{\varepsilon_\gamma} \|\hat{m} - m^\gamma\|_2 = o_P(N^{-1/2})$  before an analogous CLT can be claimed. It is not covered by the posterior-adjusted proof merely because the software can fit it.

## H.1 Regime occupation and the population ATE

Three weighting objects must be distinguished: the true stationary state probabilities  $\pi^*$ , the invariant vector  $\hat{\pi}(\hat{\Lambda})$  of a fitted transition matrix, and the empirical posterior shares  $\bar{\gamma}_N = N^{-1} \sum_t \gamma_t$ .

The implemented point estimate is  $\bar{\gamma}_N^\top \hat{\theta}_\lambda$ . With shares treated as fixed, its plug-in standard error uses  $\bar{\gamma}_N^\top \hat{\Sigma}_\lambda \bar{\gamma}_N / N$ , where  $\hat{\Sigma}_\lambda$  here denotes the long-run asymptotic covariance (the software’s `Sigma_` already includes  $1/N$ ). This fixed-share calculation is not a conditional-on-the-entire-observed-series theorem.

For the population *proxy-occupation* target  $\bar{\pi}^\top \theta_\lambda$ ,  $\bar{\pi} = \mathbb{E} \gamma_t$ , the joint influence at fixed representation is

$$\phi_{\text{occ},t} = \bar{\pi}^\top B_\lambda^{-1} \tilde{\Psi}_t + \theta_\lambda^\top (\gamma_t - \bar{\pi}).$$

HAC of this *combined* influence includes the cross-covariance between the effect and occupation estimates. This is `ate_se_` in the canonical estimator; `sate_se_` retains the fixed-share calculation. The compatibility alias `pate_se_` refers to the empirical-share calculation, not a transition-MLE ATE. The compact synthesis benchmarks retain fixed-share standard errors and report empirical coverage against simulated structural sample effects; those coverage numbers are stress-test outcomes, not guarantees.

For the different estimator  $\hat{\pi}(\hat{\Lambda})^\top \hat{\theta}_\lambda$ , let  $D_\pi = \partial \pi(\Lambda^*) / \partial \Lambda^\top$ . The separated-calibration influence additionally contributes  $(\bar{\pi}^\top B_\lambda^{-1} A_\Lambda + \theta_\lambda^\top D_\pi) \sqrt{N}(\hat{\Lambda} - \Lambda^*)$ , when  $\bar{\pi} = \pi(\Lambda^*)$ . Its variance includes both derivatives and their covariance through the calibration influence. None of the implemented empirical-share intervals estimates this transition-MLE variance.

For a *fully observed binary stationary chain*,  $\text{Cov}(S_0, S_h) = \pi_0 \pi_1 \rho^{|h|}$  gives occupation variance  $\pi_0 \pi_1 (1 + \rho) / (1 - \rho)$ . This factor applies to that occupation component only. Soft posterior occupations have their own long-run covariance, and the total ATE variance also has effect–occupation cross terms; it cannot generally be multiplied by this factor.

## I OPERATIONAL SURROGATE, SCREENING AND REGULARIZATION

**Assumption 9** (Two-sided spectral comparability). There exist  $0 < \underline{c}_J \leq \bar{c}_J < \infty$  with  $\underline{c}_J \leq \lambda_{\min}(\mathbf{J}_t) / \lambda_{\min}(\mathbf{J}) \leq \bar{c}_J$  for rolling Gram matrices  $\mathbf{J}_t$ ,  $t \geq w$ .

**Proposition 4** (Entropy surrogate). *For calibrated forward posteriors in binary regimes,  $\mathbb{E}[\|\gamma_t - \mathbf{H}_t\|_1 \mid \mathcal{F}_t] = 2 \text{Gini}(\gamma_t) \leq C_K H(\gamma_t)$  with  $C_K \leq 1/\ln 2$ . Hence  $\mathcal{D}_{\text{causal}} \equiv \varepsilon_\gamma / \lambda_{\min}(\mathbf{J}) \leq C_K \mathbb{E}[H(\gamma_t)] / \lambda_{\min}(\mathbf{J})$ , and under Assumption 9  $\mathcal{D}_{\text{causal}} \leq C_K \bar{c}_J \mathbb{E}[H(\gamma_t) / \lambda_{\min}(\mathbf{J}_t)]$ .*

The surrogate is an upper bound on  $\mathcal{D}_{\text{causal}}$ , which is defined with the *generated-state*  $\mathbf{J}$ . By Section 4 that object can be inflated relative to the oracle geometry, so the surrogate inherits phantom resolution. We keep it as a monitoring device, not a certificate. Let  $\gamma_t \equiv P(S_t \in \cdot \mid \mathcal{F}_t) \in \Delta^K$  be the Bayesian posterior state distribution

with respect to filtration  $\mathcal{F}_t \equiv \sigma(Z_{1:t})$ . Under proper Bayesian calibration, conditional on  $\mathcal{F}_t$ , the true latent indicator  $\mathbf{H}_t \equiv \mathbf{e}_{S_t}$  is distributed as a categorical random variable taking value  $\mathbf{e}_k$  with probability  $\gamma_{tk}$ .

**Step 1: Expected  $L_1$  Error as Gini Impurity.** Evaluating the conditional expectation of the  $L_1$  error:

$$\begin{aligned} \mathbb{E}[\|\gamma_t - \mathbf{H}_t\|_1 \mid \mathcal{F}_t] &= \sum_{k=1}^K \gamma_{tk} \|\gamma_t - \mathbf{e}_k\|_1 = \sum_{k=1}^K \gamma_{tk} \left( (1 - \gamma_{tk}) + \sum_{j \neq k} \gamma_{tj} \right) \\ &= \sum_{k=1}^K \gamma_{tk} \cdot 2(1 - \gamma_{tk}) = 2 \left( 1 - \sum_{k=1}^K \gamma_{tk}^2 \right) = 2 \text{Gini}(\gamma_t). \end{aligned} \quad (22)$$

**Step 2: Entropy Bound on Gini Impurity.** By the standard concave inequality  $\ln(x) \leq x - 1$ , for any  $x > 0$ :

$$H(\gamma_t) = - \sum_{k=1}^K \gamma_{tk} \ln \gamma_{tk} = \sum_{k=1}^K \gamma_{tk} \ln \left( \frac{1}{\gamma_{tk}} \right) \geq \sum_{k=1}^K \gamma_{tk} (1 - \gamma_{tk}) = \text{Gini}(\gamma_t). \quad (23)$$

Furthermore, for any binary regime partition ( $K = 2$ ) with  $\gamma_{t1} = p$  and  $\gamma_{t2} = 1 - p$ ,  $2 \text{Gini}(\gamma_t) = 4p(1 - p)$ . Comparing to binary Shannon entropy  $H(p) = -p \ln p - (1 - p) \ln(1 - p)$ , the ratio  $\frac{4p(1-p)}{H(p)}$  achieves its maximum at  $p = 1/2$ , where  $\frac{4(1/4)}{\ln 2} = \frac{1}{\ln 2} \approx 1.4427$ . Hence for all  $p \in [0, 1]$ :

$$2 \text{Gini}(\gamma_t) \leq C_K H(\gamma_t), \quad \text{where } C_K \leq \frac{1}{\ln 2} \approx 1.4427. \quad (24)$$

**Step 3: Population and Observable Bridge.** Taking the population expectation of Step 1 and applying Step 2:

$$\varepsilon_\gamma \equiv \mathbb{E}[\mathbb{E}[\|\gamma_t - \mathbf{H}_t\|_1 \mid \mathcal{F}_t]] = 2\mathbb{E}[\text{Gini}(\gamma_t)] \leq C_K \mathbb{E}[H(\gamma_t)]. \quad (25)$$

Dividing by the global minimal singular value  $\lambda_{\min}(\mathbf{J})$ :

$$\mathcal{D}_{\text{causal}} \equiv \frac{\varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})} \leq C_K \mathbb{E} \left[ \frac{H(\gamma_t)}{\lambda_{\min}(\mathbf{J})} \right] = C_K \mathbb{E}[\mathcal{D}_{\text{global}}^H(t)]. \quad (26)$$

Furthermore, under Assumption 9 (Two-Sided Spectral Comparability), the rolling Gram matrix satisfies  $\lambda_{\min}(\mathbf{J}_t) \leq \bar{c}_J \lambda_{\min}(\mathbf{J})$  almost surely for all evaluation windows  $t \geq w$ , which implies  $\frac{1}{\lambda_{\min}(\mathbf{J}_t)} \leq \frac{\bar{c}_J}{\lambda_{\min}(\mathbf{J})}$ . Pointwise substitution yields:

$$\mathcal{D}_{\text{causal}} \leq C_K \bar{c}_J \mathbb{E} \left[ \frac{H(\gamma_t)}{\lambda_{\min}(\mathbf{J}_t)} \right] = C_K \bar{c}_J \mathbb{E}[\mathcal{D}_{\text{operational}}(t)], \quad (27)$$

and  $\mathbb{E}[\mathcal{D}_{\text{operational}}(t)] \leq \bar{c}_J^{-1} \mathbb{E}[\mathcal{D}_{\text{global}}^H(t)]$ . In simulation benchmarks where true states are known and posteriors are evaluated at population parameters  $\mathbf{\Lambda}^*$ , the ergodic theorem and central limit theorem for stationary  $\alpha$ -mixing chains ensure that  $\hat{\varepsilon}_{\gamma,N} \equiv \frac{1}{N} \sum_{t=1}^N \|\gamma_t - \mathbf{H}_t\|_1 = \varepsilon_\gamma + \mathcal{O}_P(N^{-1/2})$  (preserved under  $\sqrt{N}$ -consistent estimation of  $\mathbf{\Lambda}$ ), providing an evaluation-only benchmark surrogate  $\hat{\mathcal{D}}_{\text{causal},N} \equiv \hat{\varepsilon}_{\gamma,N} / \lambda_{\min}(\mathbf{J}) = \mathcal{D}_{\text{causal}} + \mathcal{O}_P(N^{-1/2})$ . We emphasize that these bounds establish  $\mathcal{D}_{\text{operational}}(t)$  as an observable surrogate upper bound for  $\mathcal{D}_{\text{causal}}$ , not a two-sided equivalence. ■

## I.1 A Priori Proxy Diagnostics and Separation Screening

Before fitting OR-DML or downstream prediction heads, practitioners should assess proxy informativeness on auxiliary measurements  $Z_t \in \mathbb{R}^q$ . Let  $\hat{\boldsymbol{\mu}}_{Z,k} \equiv \frac{1}{N_k} \sum_{t:\hat{S}_t=k} Z_t$  and  $\hat{\boldsymbol{\Sigma}}_{Z,k}$  denote empirical cluster moments. We define the standardized separation index:

$$\Delta_Z \equiv \min_{j \neq k} \frac{\|\hat{\boldsymbol{\mu}}_{Z,j} - \hat{\boldsymbol{\mu}}_{Z,k}\|_2}{\sqrt{\frac{1}{2}(\text{Tr}(\hat{\boldsymbol{\Sigma}}_{Z,j}) + \text{Tr}(\hat{\boldsymbol{\Sigma}}_{Z,k}))/q}}. \quad (28)$$

Small fitted separation, low silhouette scores and large posterior entropy can flag weak proxy information in a chosen emission model. Mean separation alone does not characterize distributional overlap (states can differ in variances), and entropy depends on the fitted model and calibration. Thresholds such as  $\Delta_Z < 0.5$  or entropy  $> 0.80 \ln K$  are design-specific abstention heuristics, not identification tests. Theorem 2 supplies an observational-equivalence example only in its stated memoryless model class. A warning should prompt sensitivity analysis and additional identifying restrictions, not a declaration of structural non-identification.

## I.2 Spectral regularization

**Theorem 6** (Risk bound). *For  $\lambda \geq 0$  with  $\lambda_{\min}(\mathbf{J}) + \lambda > 0$ :  $\|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2 \leq (C\varepsilon_\gamma + \lambda M_\theta)/(\lambda_{\min}(\mathbf{J}) + \lambda)$ , and if  $N\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|^2$  is uniformly integrable,  $\mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 \leq 2B_\lambda^2 + 2\text{Tr}(\boldsymbol{\Omega}_\lambda^{\text{tot}})/(N(\lambda_{\min} + \lambda)^2) + o(N^{-1})$ .*

Let  $\hat{\boldsymbol{\theta}}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ , and define the population regularized estimand  $\boldsymbol{\theta}_\lambda \equiv (\mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{S}$ . We decompose the estimation error around the regularized population estimand  $\boldsymbol{\theta}_\lambda$  via the exact algebraic identity:

$$\mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 = \|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2^2 + \mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_2^2 + 2\langle \mathbb{E}[\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda], \boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^* \rangle, \quad (29)$$

which accounts explicitly for the cross term between stochastic estimation error and deterministic regularization bias. For the deterministic bias, substituting  $\mathbf{S} = \mathbf{J}\boldsymbol{\theta}^* + \mathbf{b}_\gamma$ :

$$\begin{aligned} \boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^* &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{S} - (\mathbf{J} + \lambda \mathbf{I})\boldsymbol{\theta}^*) \\ &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{S} - \mathbf{J}\boldsymbol{\theta}^* - \lambda \boldsymbol{\theta}^*) \\ &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{b}_\gamma - \lambda \boldsymbol{\theta}^*), \end{aligned} \quad (30)$$

which establishes the exact deterministic identity (30). Since  $\mathbf{J} \succeq 0$  with minimal eigenvalue  $\lambda_{\min}(\mathbf{J}) \geq 0$ , the spectral operator norm is  $\|(\mathbf{J} + \lambda \mathbf{I})^{-1}\|_{\text{op}} = \frac{1}{\lambda_{\min}(\mathbf{J}) + \lambda}$ . Applying the triangle inequality:

$$\|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2 \leq \|(\mathbf{J} + \lambda \mathbf{I})^{-1}\|_{\text{op}} (\|\mathbf{b}_\gamma\|_2 + \lambda \|\boldsymbol{\theta}^*\|_2) \leq \frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda}. \quad (31)$$

For the stochastic variance, expanding  $\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda$ :

$$\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} (\hat{\mathbf{S}} - \mathbf{S} - (\hat{\mathbf{J}} - \mathbf{J})\boldsymbol{\theta}_\lambda) + o_p(N^{-1/2}). \quad (32)$$

By Theorem 5(a),  $\sqrt{N}(\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda) \Rightarrow \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\lambda^{\text{tot}})$ . Under uniform integrability of  $N\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_2^2$ , stochastic variance satisfies  $V_N \equiv \mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_2^2 = \frac{1}{N}\text{Tr}(\boldsymbol{\Sigma}_\lambda^{\text{tot}}) + o(N^{-1}) \leq \frac{\text{Tr}(\boldsymbol{\Omega}_\lambda^{\text{tot}})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2} + o(N^{-1})$ , using  $\text{Tr}(\mathbf{A}^{-1}\boldsymbol{\Omega}\mathbf{A}^{-1}) \leq \|\mathbf{A}^{-2}\|_{\text{op}}\text{Tr}(\boldsymbol{\Omega})$  for  $\mathbf{A} = \mathbf{J} + \lambda \mathbf{I} \succ 0$ . Applying Minkowski's inequality in  $L_2(P; \mathbb{R}^K)$  to the decomposition  $(\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*) = (\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda) + (\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*)$ :

$$\left(\mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2\right)^{1/2} \leq \|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_{L_2} + \|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2 \leq \sqrt{V_N} + B_\lambda, \quad (33)$$

which establishes the finite-sample risk upper bound:

$$\mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 \leq (B_\lambda + \sqrt{V_N})^2 \leq 2B_\lambda^2 + 2V_N \leq 2\left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda}\right)^2 + \frac{2\text{Tr}(\boldsymbol{\Omega}_\lambda^{\text{tot}})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2} + o(N^{-1}), \quad (34)$$

completely avoiding any requirement to bound the stochastic bias cross-term.

**Risk Bound Minimizer vs. Empirical Grid Optimum:** Differentiating the leading-order risk upper bound  $R_{\text{bound}}(\lambda) \equiv \left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda}\right)^2 + \frac{\text{Tr}(\boldsymbol{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2}$  with respect to  $\lambda \geq 0$ , the bound-optimal shrinkage parameter  $\lambda_{\text{bound}}^* \equiv \arg \min_{\lambda \geq 0} R_{\text{bound}}(\lambda)$  satisfies:

$$\frac{\partial}{\partial \lambda} \left[ \left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda}\right)^2 + \frac{\text{Tr}(\boldsymbol{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2} \right] = 0. \quad (35)$$

When the overlap Gram matrix is ill-conditioned ( $\lambda_{\min}(\mathbf{J}) \approx 0$ ), the variance term  $\frac{\text{Tr}(\boldsymbol{\Omega})}{N(\lambda_{\min} + \lambda)^2}$  dominates for small  $\lambda$ , while the deterministic shrinkage bias  $(\frac{\lambda M_\theta}{\lambda_{\min} + \lambda})^2$  dominates for large  $\lambda$ . The interior minimizer  $\lambda_{\text{bound}}^* > 0$  balances variance reduction against shrinkage bias in the upper bound. We distinguish  $\lambda_{\text{bound}}^*$  and the empirical surrogate minimizer  $\hat{\lambda}_{\text{emp}}^* = \arg \min_\lambda \hat{R}(\lambda)$  from the true infeasible risk minimizer  $\lambda_{\text{true}}^* \equiv \arg \min_\lambda \mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2$ . In the ill-conditioned benchmark of Table 11 (100 replications, mean  $\lambda_{\min} = 0.026$ ), the grid minimum of empirical MSE is at  $\lambda = 0.1$ , a  $55\times$  reduction relative to  $\lambda = 0$ . These are properties of that DGP, not constants; legacy coverage from the earlier variance code is retired. In the well-conditioned frontier (Table 2) the penalty is immaterial.

Table 11: Regularization path in an ill-conditioned design (`reports/or_dml_regularization_frontier.csv`).

| $\lambda$ | bias   | variance | MSE    |
|-----------|--------|----------|--------|
| 0         | 0.8504 | 0.0687   | 0.7854 |
| 0.0001    | 0.8465 | 0.0684   | 0.7785 |
| 0.001     | 0.8131 | 0.0663   | 0.7210 |
| 0.005     | 0.6907 | 0.0583   | 0.5293 |
| 0.01      | 0.5784 | 0.0506   | 0.3795 |
| 0.02      | 0.4287 | 0.0395   | 0.2177 |
| 0.05      | 0.2122 | 0.0233   | 0.0615 |
| 0.1       | 0.0903 | 0.0143   | 0.0143 |
| 0.2       | 0.1259 | 0.0090   | 0.0209 |
| 0.5       | 0.3237 | 0.0048   | 0.1098 |

### I.3 Derivation of Adaptive Spectral Regularization Parameter and Oracle Bound

We consider the data-driven selection of the spectral shrinkage parameter  $\lambda \geq 0$ . While classical Tikhonov regularization selects shrinkage via prediction cross-validation, in OR-DML the target is the structural causal parameter vector  $\theta^*$ , where prediction cross-validation conflates causal identification with prognostic predictive fit. We distinguish two distinct regularization regimes:

1. **Inferential Regularization** ( $\lambda = o(N^{-1/2})$ ): Required by Theorem 5 to guarantee asymptotic unbiasedness  $\sqrt{N}\text{Bias}(\lambda) \rightarrow \mathbf{0}$ , ensuring centered Gaussian limit laws and valid nominal confidence interval coverage.
2. **Finite-Sample Risk Minimization** ( $\lambda_{\text{risk}} = \mathcal{O}(1)$ ): Minimizes the finite-sample expected  $\ell_2$  estimation risk  $\mathbb{E}\|\hat{\theta}_\lambda - \theta^*\|_2^2$  under severe causal ill-conditioning ( $\lambda_{\min}(\mathbf{J}) \approx 0$ ).

**Empirical Risk Surrogate and Convexity:** Define the empirical plug-in risk surrogate  $\hat{R}(\lambda)$  by substituting empirical estimators  $(\hat{\varepsilon}_\gamma, \lambda_{\min}(\hat{\mathbf{J}}), \hat{\Omega}, \|\hat{\theta}_0\|_2)$  into the risk upper bound of Theorem 6:

$$\hat{R}(\lambda) = \frac{(C\hat{\varepsilon}_\gamma + \lambda\|\hat{\theta}_0\|_2)^2 + N^{-1}\text{Tr}(\hat{\Omega})}{(\lambda_{\min}(\hat{\mathbf{J}}) + \lambda)^2}. \quad (36)$$

Let  $u = \lambda_{\min}(\hat{\mathbf{J}}) + \lambda \geq \lambda_{\min}(\hat{\mathbf{J}}) > 0$ . Rewriting  $\lambda = u - \lambda_{\min}$ , the numerator becomes:

$$(C\hat{\varepsilon}_\gamma - \lambda_{\min}(\hat{\mathbf{J}})\|\hat{\theta}_0\|_2 + u\|\hat{\theta}_0\|_2)^2 + N^{-1}\text{Tr}(\hat{\Omega}) = (A + uB)^2 + V, \quad (37)$$

where  $A = C\hat{\varepsilon}_\gamma - \lambda_{\min}(\hat{\mathbf{J}})\|\hat{\theta}_0\|_2$ ,  $B = \|\hat{\theta}_0\|_2$ , and  $V = N^{-1}\text{Tr}(\hat{\Omega})$ . Dividing by  $u^2$ :

$$\hat{R}(u - \lambda_{\min}) = B^2 + \frac{2AB}{u} + \frac{A^2 + V}{u^2}. \quad (38)$$

Differentiating with respect to  $u$ :

$$\frac{d\hat{R}}{du} = -\frac{2AB}{u^2} - \frac{2(A^2 + V)}{u^3} = -\frac{2}{u^3} [ABu + (A^2 + V)]. \quad (39)$$

Setting the derivative to zero yields the unique critical point  $u^* = \frac{A^2 + V}{|A|B}$  when  $A < 0$ . When  $A \geq 0$  (i.e.,  $C\hat{\varepsilon}_\gamma \geq \lambda_{\min}(\hat{\mathbf{J}})\|\hat{\theta}_0\|_2$ ),  $\frac{d\hat{R}}{du} < 0$  on  $(0, \infty)$ , so that the objective is strictly decreasing toward  $B^2$  as  $\lambda \rightarrow \infty$ . Constraining the search to a compact admissible set  $\lambda \in [0, \bar{\lambda}]$  with a fixed upper bound  $\bar{\lambda} < \infty$  (independent of  $N$ ), the bounded empirical minimizer is:

$$\hat{\lambda}^* = \begin{cases} \text{clip} \left( \frac{(C\hat{\varepsilon}_\gamma - \lambda_{\min}(\hat{\mathbf{J}})\|\hat{\theta}_0\|_2)^2 + N^{-1}\text{Tr}(\hat{\Omega})}{(\lambda_{\min}(\hat{\mathbf{J}})\|\hat{\theta}_0\|_2 - C\hat{\varepsilon}_\gamma)\|\hat{\theta}_0\|_2} - \lambda_{\min}(\hat{\mathbf{J}}), 0, \bar{\lambda} \right), & \text{if } A < 0, \\ \bar{\lambda}, & \text{if } A \geq 0. \end{cases} \quad (40)$$

Because  $R'(u) = -\frac{2}{u^3} [ABu + (A^2 + V)]$  changes sign strictly from negative to positive at  $u^*$  when  $A < 0$ ,  $u^*$  is the unique global interior minimizer on  $(0, \infty)$ .

**Lemma 4** (Oracle Inequality for the Surrogate). *Let  $R_{\text{sur}} be the population surrogate,  $\hat{R}$  its plug-in, and  $\Delta_N = \sup_{\lambda \in [0, \bar{\lambda}]} |\hat{R}(\lambda) - R_{\text{sur}}(\lambda)| = \mathcal{O}_P(N^{-1/2})$ . Then  $R_{\text{sur}}(\hat{\lambda}^*) \leq \inf_{\lambda \in [0, \bar{\lambda}]} R_{\text{sur}}(\lambda) + 2\Delta_N$ .$*

*Proof.* For any  $\lambda^\circ \in [0, \bar{\lambda}]$ :  $R_{\text{sur}}(\hat{\lambda}^*) \leq \hat{R}(\hat{\lambda}^*) + \Delta_N \leq \hat{R}(\lambda^\circ) + \Delta_N \leq R_{\text{sur}}(\lambda^\circ) + 2\Delta_N$ .  $\square$

This guarantee governs the surrogate, not the true risk; calibrated plug-in  $\hat{\varepsilon}_\gamma = \frac{2}{N} \sum_t \text{Gini}(\gamma_t)$  preserves ranking when  $C\hat{\varepsilon}_\gamma$  dominates.  $\blacksquare$

## J ALGORITHM AND SCALABILITY

Algorithm 1 (main text) lists the steps. While our primary experiments study binary regimes ( $K = 2$ ), the OR-DML coupled estimator generalizes directly to arbitrary finite state dimensions  $K \geq 2$ . Here we formalize the computational and statistical scalability of the coupled Gram matrix  $\hat{\mathbf{J}} \in \mathbb{R}^{K \times K}$ .

**Computational Complexity and Matrix Concentration under  $\alpha$ -Mixing:** Constructing  $\hat{\mathbf{J}}$  and score  $\hat{\mathbf{S}}$  requires evaluating  $K$  nuisance pairs across  $B$  folds ( $\mathcal{O}(BK \cdot \text{Cost}(\text{Nuisance}))$ ). Forming  $\hat{\mathbf{J}}$  takes  $\mathcal{O}(NK^2)$  operations, and solving  $(\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$  via Cholesky factorization takes  $\mathcal{O}(K^3)$  operations, taking  $< 1$  ms for  $K \leq 20$ . Statistically, the empirical Gram matrix  $\hat{\mathbf{J}} = \frac{1}{N} \sum_{t=1}^N \mathbf{\Gamma}_t \tilde{\mathbf{T}}_t \tilde{\mathbf{T}}_t^\top \mathbf{\Gamma}_t$  concentrates around  $\mathbf{J}$  under geometric  $\alpha$ -mixing with spectral gap  $\rho_{\text{gap}} \equiv 1 - |\lambda_2(\mathbf{\Pi})| > 0$  via the Matrix Bernstein inequality (Banna et al., 2016):

$$\|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}} \leq C_J \sqrt{\frac{K \log(K/\delta)}{N \cdot \rho_{\text{gap}}}} + \frac{C'_J K \log(K/\delta)}{N \cdot \rho_{\text{gap}}}. \quad (41)$$

By Weyl's theorem,  $|\lambda_{\min}(\hat{\mathbf{J}}) - \lambda_{\min}(\mathbf{J})| \leq \|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}}$ , ensuring that  $\lambda_{\min}(\hat{\mathbf{J}})$  concentrates at the standard  $\mathcal{O}_P(N^{-1/2})$  rate whenever  $\lambda_{\min}(\mathbf{J}) > 0$ .

**Non-Identifiability vs. Shrinkage: When Systems Must Abstain:** Spectral shrinkage  $\hat{\boldsymbol{\theta}}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$  stabilizes variance under ill-conditioned overlap ( $\lambda_{\min} \approx 0$ ). However, when data lack identifying variation entirely—diagnosed by condition number  $\kappa(\hat{\mathbf{J}}) > 10^4$  or a moving-block bootstrap 95% confidence interval for  $\lambda_{\min}(\hat{\mathbf{J}})$  that covers zero—the structural parameter  $\boldsymbol{\theta}^*$  suffers catastrophic ill-conditioning (and structural non-identification in the exact limit  $\lambda_{\min} = 0$ , Theorem 2). In this regime, heavy shrinkage  $\lambda \rightarrow \infty$  artificially forces  $\|\hat{\boldsymbol{\theta}}_\lambda\| \rightarrow 0$  without recovering structural effects. Deployed decision pipelines should treat  $\kappa(\hat{\mathbf{J}}) > 10^4$  as an operational abstention trigger rather than masking severe ill-conditioning with excessive shrinkage. Table 12 reports OR-DML with oracle-quality proxies for  $K \in \{2, 3, 4, 5\}$  ( $N = 1200$ ).

Table 12: Multi-regime scalability (reports/multiregime\_scalability\_summary.csv).

| $K$ | ATE bias | $\ \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\ _2$ | $\lambda_{\min}$ | $\kappa$ |
|-----|----------|---------------------------------------------------------|------------------|----------|
| 2   | 0.062    | 0.112                                                   | 0.758            | 1.31     |
| 3   | 0.074    | 0.142                                                   | 0.506            | 1.22     |
| 4   | 0.036    | 0.193                                                   | 0.361            | 1.43     |
| 5   | 0.061    | 0.389                                                   | 0.238            | 2.00     |

## K FACTORIAL AND LEARNED-POSTERIOR EXPERIMENTS

**Factorial design** (src/factorial\_reliability\_experiment.py). A two-state chain with persistence 0.88 and exit probability 0.15 generates  $T = 2(1-S) + 5S + 0.8X_0 - 0.5X_1 + \sigma_S V$  with  $\sigma_0 = 1$  and  $\sigma_1 \in \{0.08, \dots, 1.5\}$ , and  $Y = \theta_S T + 15(1-S) + 50S + 1.2X_0 + 0.9X_2 + U$  with  $\theta = (0.75, 2.5)$  and  $N = 1200$ . Nuisances are cross-fitted with oracle regime weights; the proxy is the label mixture  $\boldsymbol{\gamma} = (1 - \varepsilon_\gamma/2)\mathbf{H} + (\varepsilon_\gamma/2)(\mathbf{1} - \mathbf{H})$  for  $\varepsilon_\gamma \in \{0, \dots, 0.4\}$ , with 100 sequences per cell (4,900 runs). Rank correlations: run-level 0.995 for  $\varepsilon_\gamma/\lambda_{\min}$ , 0.626 for  $\varepsilon_\gamma$ , 0.521 for  $1/\lambda_{\min}$ , 0.535 for the additive score; cell-level 0.996. Retaining the 40% of runs with lowest  $\varepsilon_\gamma/\lambda_{\min}$  reduces mean error from 48.81 to 2.61. Because the proxy is a deterministic corruption of the true state, these numbers verify Proposition 1 rather than a statistical property of learned representations. Rerunning the script reproduces the committed raw data to within  $10^{-12}$ .

**Numerator scale.** In a feature-branch exploration with proxy error fixed at 0.008 and  $\lambda_{\min}$  essentially flat (0.44–0.50), increasing the treatment shift from 0.5 to 8 raises the causal error from 0.18 to 1.89. The ratio  $\varepsilon_\gamma/\lambda_{\min}$  omits the regime-shift scale that multiplies state error in both laws.

**Learned posteriors** (`src/synthesis/learned_posterior_coupled.py`). Same structural equations, with treatment shift  $\Delta_T \in \{1.5, 3, 6\}$ ,  $\sigma_1 \in \{0.3, 0.75, 1.5\}$  and 2-d Gaussian proxies with Mahalanobis separation  $\Delta_Z \in \{0.75, 1.25, 2, 3\}$ ; full-covariance HMM (three restarts, smoothing); posterior-weighted Ridge nuisances on five purged folds (embargo 12); 20 replications per cell. The oracle Gram uses the true one-hot weights with the same nuisance procedure. The plug-in shifts  $\hat{\Delta}_T, \hat{\Delta}_Y$  are posterior-weighted differences of regime means of  $T$  and  $Y$ . Raw output: `reports/synthesis/learned_posterior_coupled_raw.csv`.

## L SYNTHETIC FRONTIERS

**Observability and persistence frontier** (`src/synthesis/frontier_designs.py`). The DGP is the difficulty-frontier generator (`src/synthetic_dgp_benchmark.py`): two regimes,  $\theta = (0.75, 2.5)$ , treatment shift 4, outcome offsets 15/50, AR(1) controls, 2-d Gaussian proxies with separation  $\Delta_Z$ ,  $N = 1200$ , stationary ATE 1.53. Posteriors are learned by a full-covariance Gaussian HMM (EM, three restarts, smoothing) from  $Z$  alone. All eight estimators share the posteriors, five purged blocks (embargo 12), Ridge nuisances and a HAC lag of 12. Labels are aligned to the truth only to evaluate regime-specific errors. The grid is  $\Delta_Z \in \{0.2, 0.5, 1, 2, 4\}$  at regime-0 stay probability 0.88, plus stay probabilities  $\{0.70, 0.97\}$  at  $\Delta_Z \in \{1, 2\}$ , with 100 datasets per cell (900 in all). The DGP deliberately violates proxy sufficiency (the treatment carries state information), and  $\Delta_Z \leq 0.5$  is close to the non-identification boundary of Theorem 2, where every proxy-based estimator fails.

**Legacy difficulty benchmark** (`src/synthetic_dgp_benchmark.py`). The earlier eleven-method benchmark on the same DGP (Table 15) includes forward filtering, decoupled and unregularized variants, GBM proxy controls and HMM( $Z, T$ ); its “Spectral OR-DML” rows are the posterior-weighted design. HMM( $Z, T$ ), which infers the state from  $(Z, T)$ , is best at low separation because the treatment carries state information. The  $\lambda_{\min}$  column displays phantom resolution directly: residualizing on  $X$  only gives 4.95, against 0.45 for the oracle state, and at  $\Delta_Z = 0.2$  the weighted design’s generated-state value is 0.80 with  $\varepsilon_\gamma = 0.78$ . **Frontier B (size).** The legacy null-rejection percentages are retired because the earlier canonical implementation used a scalar score with a coupled bread. The proxy-target displacement in Theorem 5 still explains why even corrected sampling uncertainty cannot guarantee structural null size. **Soft versus hard under a continuous latent mixture** (`src/test_soft_vs_hard_continuous_mixture.py`). With a continuous latent mixture weight and an informative proxy, the posterior-weighted coupled estimator has mean absolute bias 3.71, hard fixed effects 2.81 and standard DML 3.83. Soft weighting wins in 0% of 100 replications; the experiment was written to exhibit an advantage of soft weighting and found none for that design. With posterior-adjusted nuisances on the same inputs, OR-DML has bias 2.32 and beats hard fixed effects in 100% of replications: the posterior helps once it is used as an adjustment feature rather than a nuisance weight.

## M REAL-DATA-CALIBRATED SEMI-SYNTHETIC BENCHMARK

`src/synthesis/semisynthetic_hourly.py`. For each city, the analysis sample consists of the hours on the corrected grid where  $\text{NO}_2$ ,  $\text{PM}_{2.5}$ , the proxies and the controls are all observed. The real missingness pattern is therefore inherited, including outages and the shorter Kolkata record. The proxies are temperature, humidity and wind; the controls are pressure, diurnal phase and lagged  $\text{NO}_2$ . A two-state diagonal Gaussian HMM (five restarts) is calibrated on the standardized proxies. In each of 30 replications the true path is drawn from that HMM’s exact smoothing distribution by forward-filtering backward-sampling, so regime durations and overlap are the calibrated ones. Then  $T = \text{NO}_2^{\text{std}} + S$  and  $Y = \theta_S T + 3S + X\beta + U$ , with  $\theta = (0.5, 1.5)$  and  $U$  an AR(1) process with coefficient 0.8. Estimators receive the calibrated posteriors, the most favourable case for posterior methods. Besides the estimators of Table 2, the “Spectral OR-DML (soft, coupled)” rows are the posterior-weighted design and “OR-DML (posterior-adjusted)” is the paper’s estimator. The bias-law column uses the heterogeneous-effects form with the observed  $\hat{v}$  and  $\widehat{\text{Var}}(\tilde{T})$  and the injected  $(\Delta_T, \Delta_g)$ . Bias-law prediction versus observed mean bias of posterior-feature DML: Delhi 0.36 vs 0.55; Mumbai 1.41 vs 1.03; Bengaluru 1.70 vs 1.51; Kolkata 2.00 vs 2.17. The residual gaps reflect restricted (linear) learners and the assumption  $\mathbb{E}[m_0(X)\text{Var}(D | F)] = 0$ , which the real  $\text{NO}_2$  series need not satisfy. The oracle-state intervals under-cover in some cities (70–93%), which we attribute to the HAC truncation at 24 hours under strongly persistent real  $\text{NO}_2$ .

## N REPRESENTATIONS, CALIBRATION AND RELIABILITY

LatentRegimeBench (`src/train_real_representation_zoo.py`, `src/calibration_intervention_zoo.py`, `src/hierarchical_reliability_regression.py`) trains a Gaussian HMM (causal filter), a GRU, a causal Transformer and a linear SSM on shifted worlds ( $N = 2000$ , five seeds), and evaluates state recovery, calibration and downstream failure discrimination of the rolling score  $\mathcal{D}_t = H(\gamma_t)/\lambda_{\min}(\mathbf{J}_t)$ . The HMM attains failure ROC-AUC 0.615. That is no better than plain entropy, and the neural encoders are near chance. Consistent with Theorem 4, the representation’s *probabilistic* quality (NLL) predicts reliability across worlds, but recalibration, which changes  $\hat{v}$  and not the conditional information, does not improve it.

## O SENSOR DATA, AUDIT AND FULL CITY RESULTS

**Sources.** Hourly pollutant concentrations come from CPCB continuous monitoring stations via OpenAQ (OpenAQ, 2025; CPCB, 2025), and weather from Open-Meteo reanalysis (Open-Meteo, 2025). The record runs from February 2025 to June 2026 (Kolkata to June 2025), and raw files are in `data/raw_hourly`. **Time-grid audit.** OpenAQ stamps readings at  $hh:30Z$ . The legacy pipeline (`src/data_pipeline_clean.py`) rounded these with ties-to-even, so every reading mapped to an even hour, and the inner join with hourly weather kept only even hours. The legacy “hourly” panel is in fact two-hourly: lag-1 is two hours and its rolling 3-hour mean covers six. The corrected grid (`src/bias_law/ext_real_hourly_fix.py`) floors readings to the hour, pairing each reading with the weather hour that began 30 minutes earlier (a causal alignment). It reindexes to a regular hourly grid, forward-fills at most two hours, and rebuilds lags on the grid. At least 93% of consecutive analysis rows are exactly one hour apart in every city. All real-data results in the main text use the corrected grid. Legacy two-hourly results are retained in `reports/` for provenance only. **Aligned placebos.** For a lead  $h < 0$  we regress  $\text{PM}_{2.5,t+h}$  on  $\text{NO}_{2,t}$  with the controls and contemporaneous  $\text{NO}_2$  measured at the outcome time  $t + h$ , so the test asks whether future  $\text{NO}_2$  predicts past  $\text{PM}_{2.5}$  beyond the information available then. Without this alignment, the  $h = -1$  placebo is identically zero because  $\text{PM}_{2.5,t-1}$  is a control.

## P REPRODUCTION AND VERIFICATION

**One command.** `bash reproduce.sh` reruns, in order:

1. the synthesis experiments, about ten minutes on two CPU cores: the closed-form verification, the coupled-law and orthogonality checks, the observability–persistence frontier, learned posteriors, the semi-synthetic benchmark, the city stress test, the HAC lag sensitivity and soft-versus-hard;
2. the factorial experiment (about 20 s);
3. the figures, the generated tables and `numbers.tex`;
4. `verify_submission.py`.

`bash reproduce.sh --full` also reruns the longer legacy benchmarks: the difficulty frontier, the bias-law suite and the representation zoo. **Single source of numbers.** Every numeric claim in the main text is a  $\text{\LaTeX}$  macro defined in `paper/generated/numbers.tex`, and every results table is an `\input` of `paper/generated/tab_*.tex`. Both are written by `src/synthesis/make_tables_numbers.py` from committed CSV/JSON outputs. `paper/generated/claims.csv` lists each macro with its value, source file and derivation. **Verifier.** `verify_submission.py` (i) regenerates all macros and tables into a temporary directory and requires byte-identical output, (ii) re-evaluates the closed form and its symbolic identities, (iii) checks that every macro used in the manuscript is defined and every defined macro is used or listed, (iv) checks that the PDF is newer than its sources and has an eight-page main body, and (v) scans the PDF and the supplementary archive for de-anonymizing strings.

## Q LINEAGE, CORRECTIONS AND SCOPE

This paper consolidates several development lines (an applied study, a regime-conditional DML draft, an overlap-aware DML draft, a research ledger on task-relative reliability, and a residual-state bias-law analysis). Audits

across them changed the claims as follows. (1) The two-hour time grid (Appendix O) invalidated “hourly” labels, horizons and lag lengths in earlier drafts; all real-data numbers here are recomputed. (2) Earlier placebo leads were not aligned to the outcome’s information set; the one-hour lead was mechanically zero. (3) Earlier drafts presented  $\varepsilon_\gamma/\lambda_{\min}$  as a general reliability predictor on the basis of the factorial design; Proposition 1 shows that design’s correlation is algebraic, and learned posteriors reduce it to 0.517. (4) Earlier drafts used posterior-*weighted* regime nuisances in the coupled estimator and presented it as the method of choice; every benchmark here favours hard fixed effects over that design. Proposition 1 isolates soft-weight leakage with oracle regime nuisances; Lemma 1 establishes nuisance orthogonality for posterior adjustment, and that design of Lemma 1 replaces it. (5) A semi-synthetic benchmark built on the legacy grid, in which the true state was a deterministic function of the proxies, is superseded by Appendix M. (6) A synthetic financial transfer experiment, an imputation-sensitivity table and a multi-horizon failure-forecasting table built on the legacy grid, and an early policy simulation with an unsupported mortality scaling, are not part of this paper’s evidence; their code remains in the archive for transparency. (7) Regularization numbers are recomputed from the committed path (Table 11). The full research ledger, earlier manuscripts and changelogs are in the supplementary archive under **archive/** and **research/**.

Table 13: Observability  $\times$  persistence frontier, all estimators (reports/synthesis/frontier\_designs\_summary.csv). Coverage: nominal 95% intervals for the sample ATE.  $\|\hat{\theta} - \theta^*\|$ : median over datasets.  $\lambda_{\min}$ : generated-state Gram matrix (median).

| $\Delta_Z$ | $p_{00}$ | Estimator                   | mean  bias | median | RMSE  | cover. | $\ \hat{\theta} - \theta^*\ $ | $\lambda_{\min}$ |
|------------|----------|-----------------------------|------------|--------|-------|--------|-------------------------------|------------------|
| 0.2        | 0.88     | Oracle-state                | 0.029      | 0.027  | 0.035 | 95%    | 0.064                         | 0.435            |
|            |          | Standard DML                | 8.442      | 8.445  | 8.442 | 0%     | —                             | —                |
|            |          | Block DML                   | 8.472      | 8.470  | 8.473 | 0%     | —                             | —                |
|            |          | DML + Z controls            | 8.291      | 8.292  | 8.292 | 0%     | —                             | —                |
|            |          | Posterior-feature DML       | 8.162      | 8.156  | 8.165 | 0%     | —                             | —                |
|            |          | Hard regime FE              | 8.240      | 8.231  | 8.242 | 0%     | 11.347                        | 1.076            |
|            |          | OR-DML, weighted nuisances  | 8.634      | 8.646  | 8.636 | 0%     | 12.144                        | 0.460            |
|            |          | OR-DML (posterior-adjusted) | 8.126      | 8.115  | 8.129 | 0%     | 11.230                        | 0.404            |
| 0.5        | 0.88     | Oracle-state                | 0.028      | 0.024  | 0.034 | 95%    | 0.058                         | 0.451            |
|            |          | Standard DML                | 8.421      | 8.418  | 8.422 | 0%     | —                             | —                |
|            |          | Block DML                   | 8.455      | 8.456  | 8.455 | 0%     | —                             | —                |
|            |          | DML + Z controls            | 7.427      | 7.408  | 7.428 | 0%     | —                             | —                |
|            |          | Posterior-feature DML       | 6.009      | 5.996  | 6.014 | 0%     | —                             | —                |
|            |          | Hard regime FE              | 6.460      | 6.461  | 6.465 | 0%     | 9.109                         | 1.119            |
|            |          | OR-DML, weighted nuisances  | 9.025      | 9.068  | 9.029 | 0%     | 12.865                        | 0.760            |
|            |          | OR-DML (posterior-adjusted) | 5.940      | 5.937  | 5.946 | 0%     | 8.446                         | 0.554            |
| 1.0        | 0.88     | Oracle-state                | 0.026      | 0.025  | 0.033 | 96%    | 0.052                         | 0.452            |
|            |          | Standard DML                | 8.432      | 8.444  | 8.433 | 0%     | —                             | —                |
|            |          | Block DML                   | 8.465      | 8.463  | 8.466 | 0%     | —                             | —                |
|            |          | DML + Z controls            | 5.433      | 5.440  | 5.435 | 0%     | —                             | —                |
|            |          | Posterior-feature DML       | 2.164      | 2.080  | 2.280 | 0%     | —                             | —                |
|            |          | Hard regime FE              | 2.575      | 2.489  | 2.674 | 0%     | 3.652                         | 0.597            |
|            |          | OR-DML, weighted nuisances  | 4.236      | 4.210  | 4.282 | 0%     | 6.107                         | 0.547            |
|            |          | OR-DML (posterior-adjusted) | 2.134      | 2.046  | 2.252 | 0%     | 2.954                         | 0.498            |
| 2.0        | 0.88     | Oracle-state                | 0.031      | 0.027  | 0.038 | 94%    | 0.060                         | 0.448            |
|            |          | Standard DML                | 8.437      | 8.437  | 8.438 | 0%     | —                             | —                |
|            |          | Block DML                   | 8.474      | 8.476  | 8.475 | 0%     | —                             | —                |
|            |          | DML + Z controls            | 2.633      | 2.640  | 2.637 | 0%     | —                             | —                |
|            |          | Posterior-feature DML       | 0.141      | 0.045  | 0.865 | 98%    | —                             | —                |
|            |          | Hard regime FE              | 0.141      | 0.042  | 0.852 | 93%    | 0.077                         | 0.453            |
|            |          | OR-DML, weighted nuisances  | 0.147      | 0.045  | 0.867 | 94%    | 0.095                         | 0.451            |
|            |          | OR-DML (posterior-adjusted) | 0.129      | 0.038  | 0.860 | 91%    | 0.070                         | 0.449            |
| 4.0        | 0.88     | Oracle-state                | 0.031      | 0.027  | 0.037 | 94%    | 0.063                         | 0.445            |
|            |          | Standard DML                | 8.444      | 8.445  | 8.444 | 0%     | —                             | —                |
|            |          | Block DML                   | 8.481      | 8.484  | 8.481 | 0%     | —                             | —                |
|            |          | DML + Z controls            | 0.864      | 0.862  | 0.870 | 0%     | —                             | —                |
|            |          | Posterior-feature DML       | 0.044      | 0.036  | 0.056 | 100%   | —                             | —                |
|            |          | Hard regime FE              | 0.031      | 0.027  | 0.037 | 94%    | 0.063                         | 0.445            |
|            |          | OR-DML, weighted nuisances  | 0.032      | 0.028  | 0.037 | 94%    | 0.063                         | 0.445            |
|            |          | OR-DML (posterior-adjusted) | 0.031      | 0.028  | 0.037 | 94%    | 0.063                         | 0.445            |
| 1.0        | 0.70     | Oracle-state                | 0.027      | 0.024  | 0.033 | 95%    | 0.064                         | 0.337            |
|            |          | Standard DML                | 7.961      | 7.945  | 7.962 | 0%     | —                             | —                |
|            |          | Block DML                   | 7.978      | 7.954  | 7.979 | 0%     | —                             | —                |
|            |          | DML + Z controls            | 5.174      | 5.183  | 5.177 | 0%     | —                             | —                |
|            |          | Posterior-feature DML       | 2.742      | 2.726  | 2.770 | 0%     | —                             | —                |
|            |          | Hard regime FE              | 3.260      | 3.230  | 3.290 | 0%     | 5.220                         | 0.546            |
|            |          | OR-DML, weighted nuisances  | 5.256      | 5.260  | 5.280 | 0%     | 8.811                         | 0.470            |
|            |          | OR-DML (posterior-adjusted) | 2.725      | 2.705  | 2.751 | 0%     | 4.467                         | 0.409            |
| 2.0        | 0.70     | Oracle-state                | 0.027      | 0.023  | 0.034 | 96%    | 0.060                         | 0.335            |
|            |          | Standard DML                | 7.959      | 7.953  | 7.960 | 0%     | —                             | —                |
|            |          | Block DML                   | 7.979      | 7.971  | 7.980 | 0%     | —                             | —                |
|            |          | DML + Z controls            | 2.521      | 2.515  | 2.524 | 0%     | —                             | —                |
|            |          | Posterior-feature DML       | 0.060      | 0.049  | 0.083 | 98%    | —                             | —                |
|            |          | Hard regime FE              | 0.070      | 0.039  | 0.101 | 97%    | 0.084                         | 0.336            |
|            |          | OR-DML, weighted nuisances  | 0.079      | 0.048  | 0.112 | 94%    | 0.123                         | 0.335            |
|            |          | OR-DML (posterior-adjusted) | 0.052      | 0.032  | 0.073 | 96%    | 0.076                         | 0.336            |
| 1.0        | 0.97     | Oracle-state                | 0.025      | 0.021  | 0.033 | 93%    | 0.078                         | 0.175            |
|            |          | Standard DML                | 7.609      | 7.641  | 7.616 | 0%     | —                             | —                |
|            |          | Block DML                   | 7.679      | 7.677  | 7.686 | 0%     | —                             | —                |
|            |          | DML + Z controls            | 5.228      | 5.244  | 5.231 | 0%     | —                             | —                |
|            |          | Posterior-feature DML       | 1.060      | 1.072  | 1.114 | 1%     | —                             | —                |
|            |          | Hard regime FE              | 1.194      | 1.182  | 1.258 | 3%     | 2.483                         | 0.225            |
|            |          | OR-DML, weighted nuisances  | 1.933      | 1.956  | 1.976 | 0%     | 5.088                         | 0.211            |
|            |          | OR-DML (posterior-adjusted) | 0.960      | 0.933  | 1.010 | 0%     | 2.282                         | 0.197            |
| 2.0        | 0.97     | Oracle-state                | 0.027      | 0.024  | 0.034 | 93%    | 0.074                         | 0.159            |
|            |          | Standard DML                | 7.486      | 7.530  | 7.496 | 0%     | —                             | —                |
|            |          | Block DML                   | 7.556      | 7.577  | 7.567 | 0%     | —                             | —                |
|            |          | DML + Z controls            | 2.639      | 2.632  | 2.642 | 0%     | —                             | —                |
|            |          | Posterior-feature DML       | 0.051      | 0.039  | 0.069 | 97%    | —                             | —                |
|            |          | Hard regime FE              | 0.039      | 0.025  | 0.059 | 94%    | 0.087                         | 0.159            |
|            |          | OR-DML, weighted nuisances  | 0.041      | 0.028  | 0.059 | 94%    | 0.098                         | 0.159            |
|            |          | OR-DML (posterior-adjusted) | 0.035      | 0.024  | 0.051 | 93%    | 0.086                         | 0.159            |

Table 14: Persistence: mean absolute ATE bias by regime-0 stay probability  $p_{00}$  (reports/synthesis/frontier\_designs\_summary.csv).

| $p_{00}$ | $\Delta_Z$ | switches | $\varepsilon_\gamma$ | Post.-feature | Hard FE | OR-DML wtd. | <b>OR-DML</b> | oracle |
|----------|------------|----------|----------------------|---------------|---------|-------------|---------------|--------|
| 0.70     | 1          | 239      | 0.090                | 2.742         | 3.260   | 5.256       | 2.725         | 0.027  |
| 0.70     | 2          | 243      | 0.001                | 0.060         | 0.070   | 0.079       | 0.052         | 0.027  |
| 0.88     | 1          | 158      | 0.062                | 2.164         | 2.575   | 4.236       | 2.134         | 0.026  |
| 0.88     | 2          | 159      | 0.000                | 0.141         | 0.141   | 0.147       | 0.129         | 0.031  |
| 0.97     | 1          | 62       | 0.026                | 1.060         | 1.194   | 1.933       | 0.960         | 0.025  |
| 0.97     | 2          | 58       | 0.000                | 0.051         | 0.039   | 0.041       | 0.035         | 0.027  |

Table 15: Legacy difficulty benchmark, all methods (reports/or\_dml\_benchmark\_summary.csv). Legacy uncertainty columns are omitted because they predate the corrected variance implementation.

| $\Delta_Z$ | Estimator                 | mean  bias | median | RMSE  | $\lambda_{\min}$ |
|------------|---------------------------|------------|--------|-------|------------------|
| 0.2        | Oracle DML                | 0.056      | 0.044  | 0.070 | 0.449            |
|            | Standard DML              | 8.456      | 8.456  | 8.456 | 4.934            |
|            | Block DML                 | 8.490      | 8.479  | 8.491 | 5.011            |
|            | DML + Z Controls (Ridge)  | 8.307      | 8.312  | 8.308 | 4.627            |
|            | DML + Z Controls (GBM)    | 8.018      | 8.025  | 8.019 | 5.325            |
|            | Regime FE DML (Hard)      | 8.318      | 8.324  | 8.319 | 1.745            |
|            | Unregularized Coupled DML | 8.637      | 8.615  | 8.638 | 0.804            |
|            | Spectral OR-DML           | 8.627      | 8.604  | 8.628 | 0.804            |
|            | Filtered OR-DML           | 8.604      | 8.582  | 8.605 | 0.800            |
|            | Decoupled Soft OR-DML     | 8.381      | 8.381  | 8.381 | 1.534            |
| 0.5        | HMM( $Z, T$ ) OR-DML      | 6.504      | 6.606  | 6.546 | 0.487            |
|            | Oracle DML                | 0.055      | 0.044  | 0.071 | 0.448            |
|            | Standard DML              | 8.429      | 8.432  | 8.430 | 4.956            |
|            | Block DML                 | 8.467      | 8.483  | 8.468 | 5.041            |
|            | DML + Z Controls (Ridge)  | 7.429      | 7.424  | 7.430 | 3.401            |
|            | DML + Z Controls (GBM)    | 7.031      | 7.020  | 7.034 | 3.819            |
|            | Regime FE DML (Hard)      | 6.628      | 6.672  | 6.634 | 1.203            |
|            | Unregularized Coupled DML | 8.848      | 8.887  | 8.851 | 0.828            |
|            | Spectral OR-DML           | 8.835      | 8.874  | 8.839 | 0.828            |
|            | Filtered OR-DML           | 8.988      | 9.003  | 8.991 | 0.875            |
| 1.0        | Decoupled Soft OR-DML     | 7.164      | 7.178  | 7.167 | 1.311            |
|            | HMM( $Z, T$ ) OR-DML      | 4.324      | 4.390  | 4.364 | 0.482            |
|            | Oracle DML                | 0.057      | 0.043  | 0.073 | 0.452            |
|            | Standard DML              | 8.433      | 8.449  | 8.434 | 4.954            |
|            | Block DML                 | 8.466      | 8.475  | 8.467 | 5.027            |
|            | DML + Z Controls (Ridge)  | 5.438      | 5.462  | 5.440 | 2.078            |
|            | DML + Z Controls (GBM)    | 3.633      | 3.642  | 3.647 | 1.885            |
|            | Regime FE DML (Hard)      | 2.647      | 2.700  | 2.685 | 0.609            |
|            | Unregularized Coupled DML | 4.025      | 4.030  | 4.045 | 0.557            |
|            | Spectral OR-DML           | 4.017      | 4.022  | 4.037 | 0.557            |
| 2.0        | Filtered OR-DML           | 5.172      | 5.185  | 5.185 | 0.605            |
|            | Decoupled Soft OR-DML     | 3.294      | 3.286  | 3.310 | 0.670            |
|            | HMM( $Z, T$ ) OR-DML      | 1.196      | 1.192  | 1.230 | 0.460            |
|            | Oracle DML                | 0.052      | 0.042  | 0.066 | 0.454            |
|            | Standard DML              | 8.432      | 8.451  | 8.433 | 4.954            |
|            | Block DML                 | 8.461      | 8.472  | 8.462 | 5.024            |
|            | DML + Z Controls (Ridge)  | 2.631      | 2.614  | 2.635 | 1.351            |
|            | DML + Z Controls (GBM)    | 0.207      | 0.194  | 0.243 | 1.214            |
|            | Regime FE DML (Hard)      | 0.077      | 0.063  | 0.102 | 0.455            |
|            | Unregularized Coupled DML | 0.082      | 0.073  | 0.105 | 0.455            |
| 4.0        | Spectral OR-DML           | 0.081      | 0.072  | 0.104 | 0.455            |
|            | Filtered OR-DML           | 0.102      | 0.081  | 0.131 | 0.456            |
|            | Decoupled Soft OR-DML     | 0.079      | 0.069  | 0.101 | 0.456            |
|            | HMM( $Z, T$ ) OR-DML      | 0.059      | 0.045  | 0.074 | 0.454            |
|            | Oracle DML                | 0.054      | 0.048  | 0.067 | 0.439            |
|            | Standard DML              | 8.451      | 8.461  | 8.452 | 4.912            |
|            | Block DML                 | 8.486      | 8.495  | 8.487 | 4.996            |
|            | DML + Z Controls (Ridge)  | 0.848      | 0.842  | 0.853 | 1.095            |
|            | DML + Z Controls (GBM)    | 0.101      | 0.082  | 0.125 | 1.189            |
|            | Regime FE DML (Hard)      | 0.054      | 0.048  | 0.067 | 0.439            |
| 4.0        | Unregularized Coupled DML | 0.054      | 0.048  | 0.067 | 0.439            |
|            | Spectral OR-DML           | 0.054      | 0.046  | 0.067 | 0.439            |
|            | Filtered OR-DML           | 0.054      | 0.046  | 0.067 | 0.439            |
|            | Decoupled Soft OR-DML     | 0.054      | 0.046  | 0.067 | 0.439            |
|            | HMM( $Z, T$ ) OR-DML      | 0.054      | 0.046  | 0.067 | 0.439            |
|            | Oracle DML                | 0.054      | 0.046  | 0.067 | 0.439            |

Table 16: Semi-synthetic benchmark, all estimators (`reports/synthesis/semisynthetic_hourly_summary.csv`). Coverage: nominal 95% intervals for the sample ATE.

| City      | Estimator                       | bias  | RMSE  | cover. | $\ \hat{\theta} - \theta\ $ | $\lambda_{\min}$ prx/orc |
|-----------|---------------------------------|-------|-------|--------|-----------------------------|--------------------------|
| Delhi     | Oracle-state regime DML         | 0.028 | 0.034 | 83%    | 0.05                        |                          |
|           | Standard DML (random folds)     | 1.942 | 1.942 | 0%     | –                           |                          |
|           | Block DML (purged folds)        | 2.163 | 2.163 | 0%     | –                           |                          |
|           | DML + Z controls                | 1.222 | 1.222 | 0%     | –                           |                          |
|           | Posterior-feature DML           | 0.551 | 0.552 | 0%     | –                           |                          |
|           | Hard regime FE DML              | 0.549 | 0.550 | 0%     | 0.77                        |                          |
|           | Spectral OR-DML (soft, coupled) | 0.879 | 0.880 | 0%     | 1.21                        | 0.085/0.082              |
|           | OR-DML (posterior-adjusted)     | 0.417 | 0.420 | 0%     | 0.59                        | 0.085/0.082              |
| Mumbai    | Oracle-state regime DML         | 0.073 | 0.094 | 83%    | 0.16                        |                          |
|           | Standard DML (random folds)     | 3.004 | 3.005 | 0%     | –                           |                          |
|           | Block DML (purged folds)        | 3.088 | 3.089 | 0%     | –                           |                          |
|           | DML + Z controls                | 3.027 | 3.028 | 0%     | –                           |                          |
|           | Posterior-feature DML           | 1.028 | 1.032 | 0%     | –                           |                          |
|           | Hard regime FE DML              | 1.861 | 1.863 | 0%     | 2.84                        |                          |
|           | Spectral OR-DML (soft, coupled) | 3.102 | 3.103 | 0%     | 4.59                        | 0.024/0.010              |
|           | OR-DML (posterior-adjusted)     | 1.651 | 1.653 | 0%     | 2.56                        | 0.024/0.010              |
| Bengaluru | Oracle-state regime DML         | 0.116 | 0.134 | 70%    | 0.22                        |                          |
|           | Standard DML (random folds)     | 2.971 | 2.971 | 0%     | –                           |                          |
|           | Block DML (purged folds)        | 3.203 | 3.204 | 0%     | –                           |                          |
|           | DML + Z controls                | 2.602 | 2.603 | 0%     | –                           |                          |
|           | Posterior-feature DML           | 1.513 | 1.515 | 0%     | –                           |                          |
|           | Hard regime FE DML              | 1.973 | 1.975 | 0%     | 2.83                        |                          |
|           | Spectral OR-DML (soft, coupled) | 3.219 | 3.220 | 0%     | 4.56                        | 0.019/0.011              |
|           | OR-DML (posterior-adjusted)     | 1.690 | 1.692 | 0%     | 2.40                        | 0.019/0.011              |
| Kolkata   | Oracle-state regime DML         | 0.136 | 0.179 | 93%    | 0.33                        |                          |
|           | Standard DML (random folds)     | 3.228 | 3.230 | 0%     | –                           |                          |
|           | Block DML (purged folds)        | 3.298 | 3.301 | 0%     | –                           |                          |
|           | DML + Z controls                | 3.055 | 3.058 | 0%     | –                           |                          |
|           | Posterior-feature DML           | 2.165 | 2.168 | 0%     | –                           |                          |
|           | Hard regime FE DML              | 2.300 | 2.304 | 0%     | 3.21                        |                          |
|           | Spectral OR-DML (soft, coupled) | 3.813 | 3.816 | 0%     | 5.33                        | 0.018/0.008              |
|           | OR-DML (posterior-adjusted)     | 2.053 | 2.058 | 0%     | 2.88                        | 0.018/0.008              |

Table 17: Temperature scaling (`reports/representation_zoo_calibration_intervention.csv`).

| Encoder                        | $T^*$ | ECE raw→cal.  | NLL raw→cal. | AUC( $\mathcal{D}_t$ ) raw→cal. |
|--------------------------------|-------|---------------|--------------|---------------------------------|
| Gaussian HMM (Causal Filter)   | 1.00  | 0.0416→0.0416 | 0.205→0.205  | 0.5895→0.5895                   |
| Neural GRU Encoder             | 3.20  | 0.1050→0.0569 | 0.547→0.331  | 0.5644→0.5609                   |
| Causal Transformer Encoder     | 4.80  | 0.1379→0.0541 | 1.006→0.382  | 0.5356→0.5282                   |
| Linear State Space Model (SSM) | 1.54  | 0.0496→0.0386 | 0.325→0.302  | 0.5492→0.5463                   |

Table 18: Leave-one-world-out prediction of downstream reliability (15 worlds, `reports/representation_zoo_lowo_evaluation.csv`). With 15 worlds the effective sample is small; we report the comparison, not a significance claim.

| LOWO predictors | RMSE   | $R^2$ | Spearman $\rho$ | $p$     |
|-----------------|--------|-------|-----------------|---------|
| $F_1$           | 0.0525 | 0.113 | 0.374           | 3.2e-03 |
| $F_1$ +ECE      | 0.0513 | 0.152 | 0.439           | 4.5e-04 |
| $F_1$ +NLL      | 0.0492 | 0.221 | 0.526           | 1.6e-05 |
| $F_1$ +ECE+NLL  | 0.0498 | 0.204 | 0.495           | 5.7e-05 |

Table 19: Dynamic responses and lead placebos by horizon (hours; ATE, \*:  $p < 0.05$ ; `reports/synthesis/city_placebo_hourly.csv`). Delhi’s response peaks at 1.61 at  $h = 6$ h, but its one-hour lead rejects, so we do not interpret the curve causally.

| City      | screen | $h = -6$ | $h = -3$ | $h = -1$ | $h = 0$ | $h = 1$ | $h = 3$ | $h = 6$ | $h = 12$ | $h = 24$ |
|-----------|--------|----------|----------|----------|---------|---------|---------|---------|----------|----------|
| Delhi     | no     | -0.02    | 0.02     | 0.07*    | 0.23*   | 0.70*   | 1.49*   | 1.61*   | 0.81*    | 0.33*    |
| Mumbai    | no     | -0.35    | -0.47    | 1.79     | 3.02    | 5.68*   | 2.73    | -5.46*  | -4.94    | -9.90*   |
| Bengaluru | no     | -0.03*   | -0.04*   | -0.05    | 0.00    | 0.01    | -0.01   | 0.01    | -0.05    | -0.10    |
| Kolkata   | no     | -0.04    | -0.17    | 0.91     | 1.22    | 1.05    | -0.82   | -2.71   | -2.24    | 0.30     |
| Delhi     | yes    | -0.02    | 0.02     | 0.07*    | 0.23*   | 0.70*   | 1.49*   | 1.61*   | 0.81*    | 0.33*    |
| Mumbai    | yes    | -9.28*   | 3.16     | 5.42     | 0.38    | 4.17    | -3.47   | -4.52   | -7.70    | -15.61*  |
| Bengaluru | yes    | -0.04*   | -0.05*   | -0.06    | 0.02    | 0.03    | 0.00    | -0.01   | -0.08    | -0.13*   |
| Kolkata   | yes    | 0.39     | 0.92     | 3.16*    | 3.32*   | 4.14*   | 1.77    | 3.88    | 17.20*   | 36.04*   |

Table 20: Newey–West lag sensitivity of regime-specific OR-DML with posterior-adjusted nuisances (corrected grid, frozen-sensor screen; `reports/bias_low/e6_hac_lag_sensitivity.csv`).

| City      | regime | lag 12                     | lag 24                     | lag 72                     | lag 168                    |
|-----------|--------|----------------------------|----------------------------|----------------------------|----------------------------|
| Delhi     | 1      | 0.430 (0.050; $p=0.000$ )  | 0.430 (0.053; $p=0.000$ )  | 0.430 (0.059; $p=0.000$ )  | 0.430 (0.065; $p=0.000$ )  |
|           | 2      | 0.102 (0.033; $p=0.002$ )  | 0.102 (0.033; $p=0.002$ )  | 0.102 (0.036; $p=0.005$ )  | 0.102 (0.040; $p=0.011$ )  |
| Mumbai    | 1      | -4.012 (4.967; $p=0.419$ ) | -4.012 (4.687; $p=0.392$ ) | -4.012 (4.869; $p=0.410$ ) | -4.012 (5.223; $p=0.442$ ) |
|           | 2      | 2.757 (1.473; $p=0.061$ )  | 2.757 (1.496; $p=0.065$ )  | 2.757 (1.547; $p=0.075$ )  | 2.757 (1.623; $p=0.089$ )  |
| Bengaluru | 1      | 0.182 (0.149; $p=0.222$ )  | 0.182 (0.155; $p=0.240$ )  | 0.182 (0.183; $p=0.320$ )  | 0.182 (0.196; $p=0.354$ )  |
|           | 2      | -0.158 (0.059; $p=0.008$ ) | -0.158 (0.063; $p=0.012$ ) | -0.158 (0.073; $p=0.031$ ) | -0.158 (0.085; $p=0.063$ ) |
| Kolkata   | 1      | -0.027 (0.522; $p=0.959$ ) | -0.027 (0.544; $p=0.961$ ) | -0.027 (0.617; $p=0.965$ ) | -0.027 (0.581; $p=0.963$ ) |
|           | 2      | 7.084 (3.450; $p=0.040$ )  | 7.084 (3.401; $p=0.037$ )  | 7.084 (3.279; $p=0.031$ )  | 7.084 (3.260; $p=0.030$ )  |

Table 21: Sensitivity analysis on the corrected hourly grid (posterior-feature estimate, 90% block-bootstrap intervals; `reports/bias_low/x23_real_hourly_fix.csv`).

| City      | screened | $\hat{\theta}$ [90% CI] | $b^*$ (SD) [90% CI] | $b^*/ \hat{c}_Y $ |
|-----------|----------|-------------------------|---------------------|-------------------|
| Delhi     | no       | 0.37 [0.29, 0.44]       | 82.3 [54.3, 144]    | 560               |
| Mumbai    | no       | 2.23 [-0.66, 4.89]      | 11.9 [1.3, 43]      | 392               |
| Bengaluru | no       | 0.01 [-0.11, 0.14]      | 26.8 [3.0, 445]     | 158               |
| Kolkata   | no       | 1.79 [-0.74, 4.60]      | 10.9 [1.3, 105]     | 35                |
| Mumbai    | yes      | 0.31 [-2.80, 2.88]      | 1.2 [0.5, 18]       | 39                |
| Bengaluru | yes      | 0.02 [-0.12, 0.16]      | 35.3 [3.3, 375]     | 189               |
| Kolkata   | yes      | 4.53 [1.48, 7.42]       | 16.5 [4.2, 60]      | 38                |